A Step-By-Step Guide to BirdNET Analyzer

Rebecca Cohen, Irina Tolkova, Meghan Beatty, Spencer Keyser, Isha Bopardikar, Chiti Arvind, V.V. Robin

Adapted and updated from materials previously created by Ashraft Syazwan Ahmady Yusni, Nor Amira Abdul Rahman, Maria Vasilkin, and Dena Clink, and based on: Symes L, Sugai LSMS, Gottesman B, Pitzrick M, Wood C. 2023. Acoustic analysis with BirdNET and (almost) no coding: practical instructions. https://doi.org/10.5281/zenodo.8357176

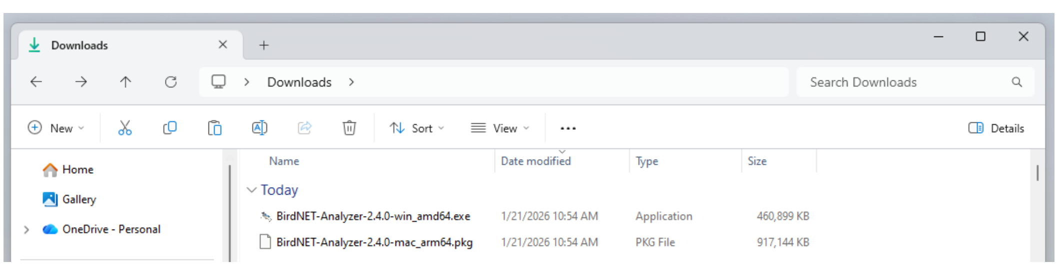
The data and downloads for this guide can be found here

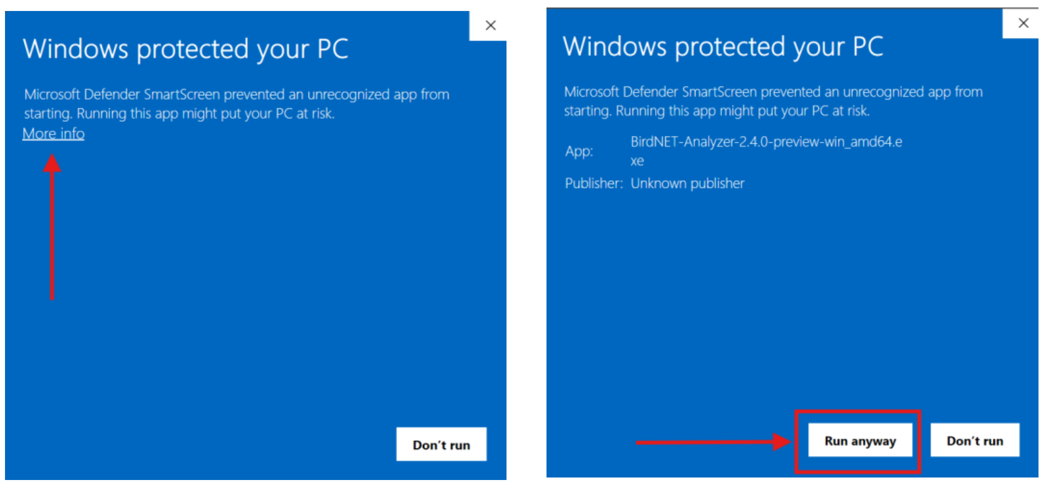
Part 1: Software Install

1.1 BirdNET Analyzer v.2.4.0

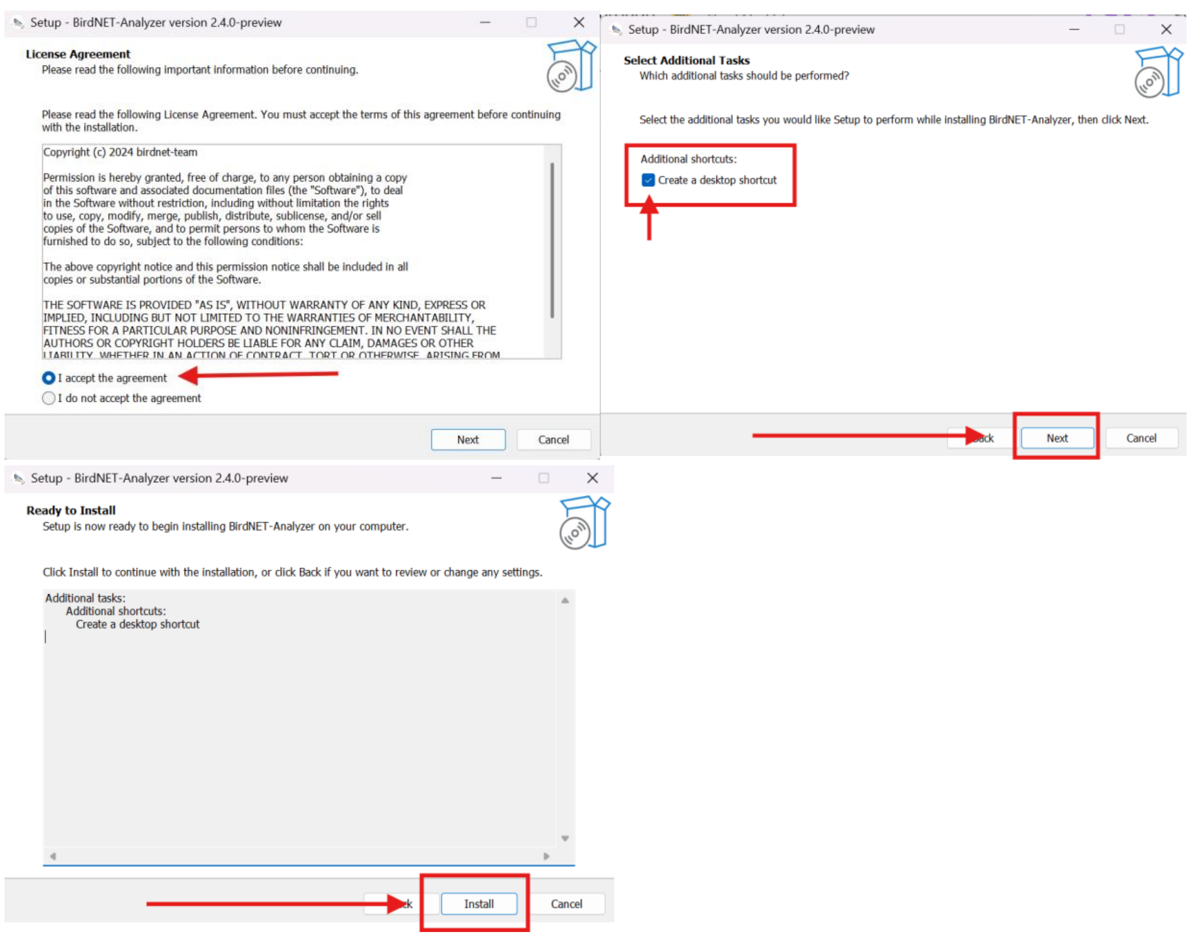
1. Download BirdNET Analyzer v2.4.0 installer from the BirdNET GitHub repository releases page
    - `BirdNET-Analyzer-2.4.0-win_amd64.exe` for Windows
    - `BirdNET-Analyzer-2.4.0-mac_arm64.pkg` for MAC

2. Double click the downloaded installer file to run it; you may need to tell your computer that it is safe to run. Click ***More info*** and ***Run anyway***.





3. Accept the license agreement and click ***Next***. Check the box to create a desktop shortcut and click ***Next***. Click ***Install***.

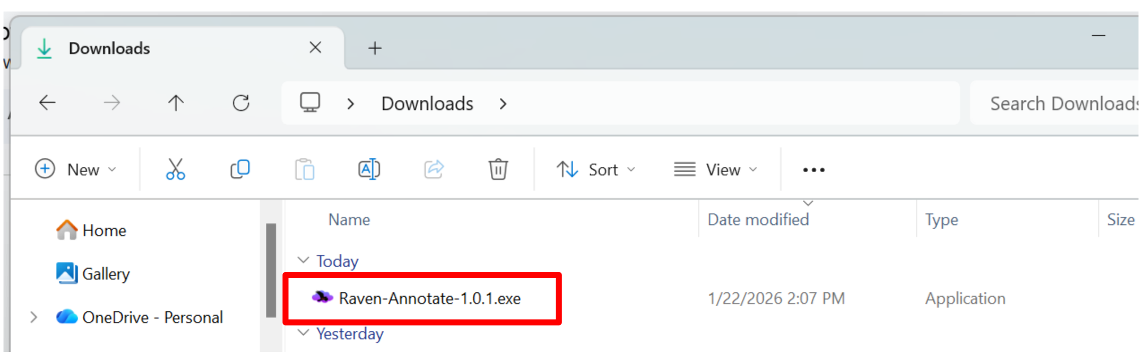


1.2 Raven Annotate Pre-Release Build (install optional)

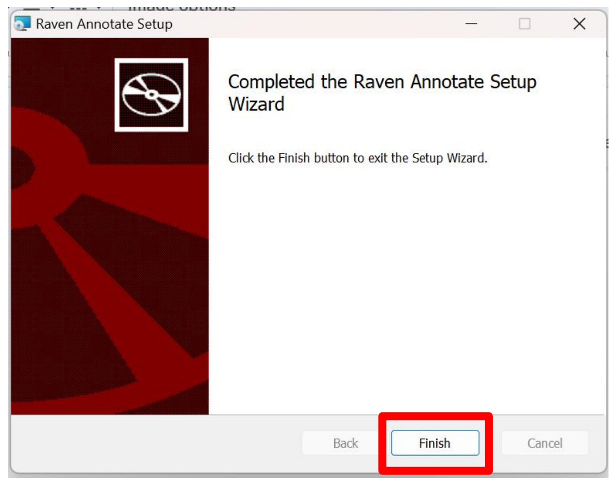
Raven Annotate is a new Raven Workbench module that is intended to replace Raven Pro's functionality for visualizing raw audio files, annotating acoustic events, making automatic feature measurements, and more. Raven Annotate is still in development and has not yet been publicly released, but the basic visualization and annotation functionality is up and running in this build. New functionality will continue to be developed and released in coming months. **If you have Raven Pro or Raven Lite already installed, it is optional to install Raven Annotate.**

1. Use one of the following links to download the appropriate Raven Annotate installer, according to your operating system:
   a. Windows:https://updates.ravensoundsoftware.com/updates/workbench/raven_annotate/win_x64/Raven-Annotate-1.1.0.exe
   b. MacOS:https://updates.ravensoundsoftware.com/updates/workbench/raven_annotate/mac_arm64/Raven-Annotate-1.1.0.dmg
   c. Linux:https://updates.ravensoundsoftware.com/updates/workbench/raven_annotate/linux_x64/raven-annotate_1.1.0-1_amd64.deb

2. Locate the installer in your Downloads folder and double click to run.



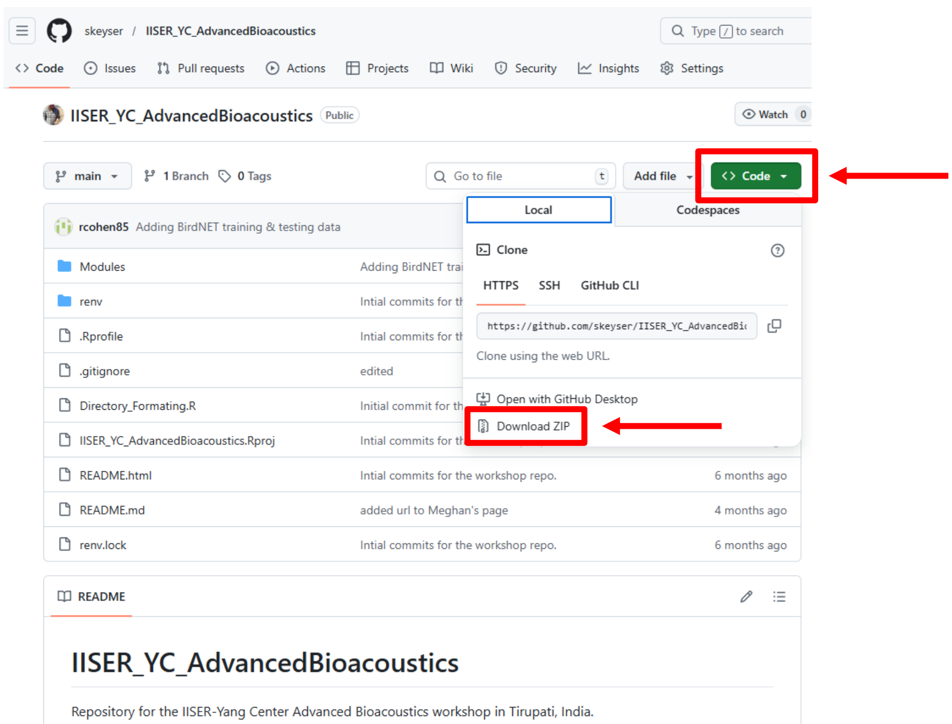
3. Follow the prompts to complete installation by accepting the license agreement and clicking ***Install***.



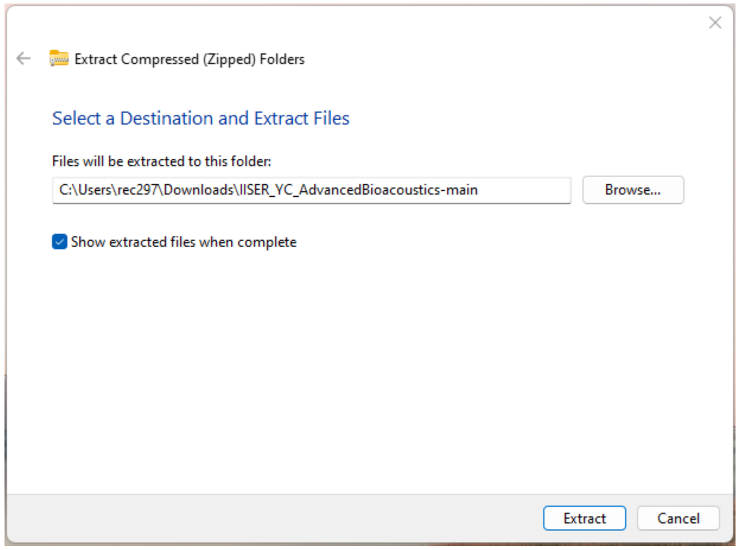
Part 2: Detecting with BirdNET Model V2.4

2.1 Download workshop materials

1. From the GitHub page, **download workshop repository "IISER_YC_AdvancedBioacoustics" as a zip file.** This repository contains all of the data and code we will be using for the modules we'll cover this week. Slide decks will be added at the end of the workshop.



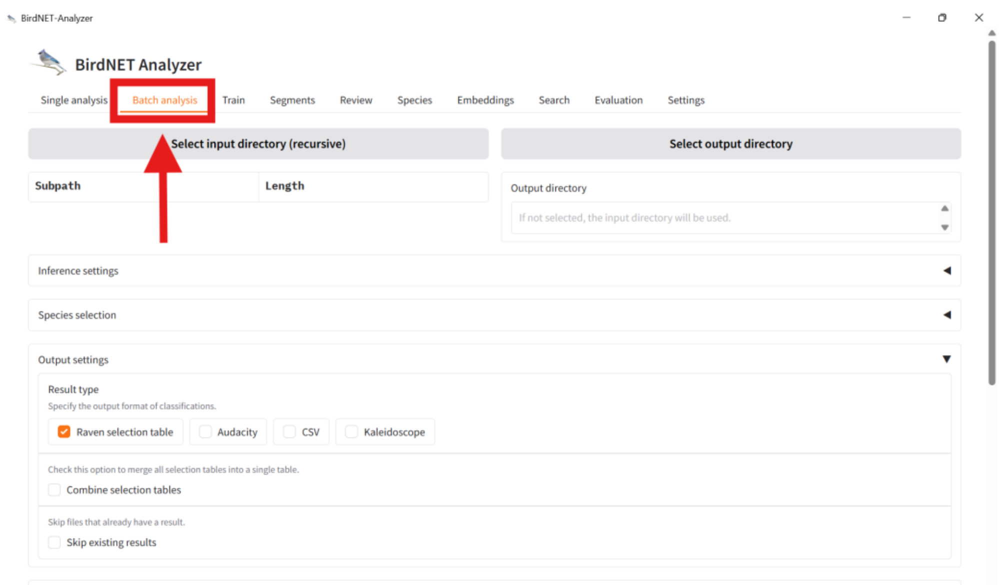
2. **Unzip the folder** and store it in a convenient location on your computer.



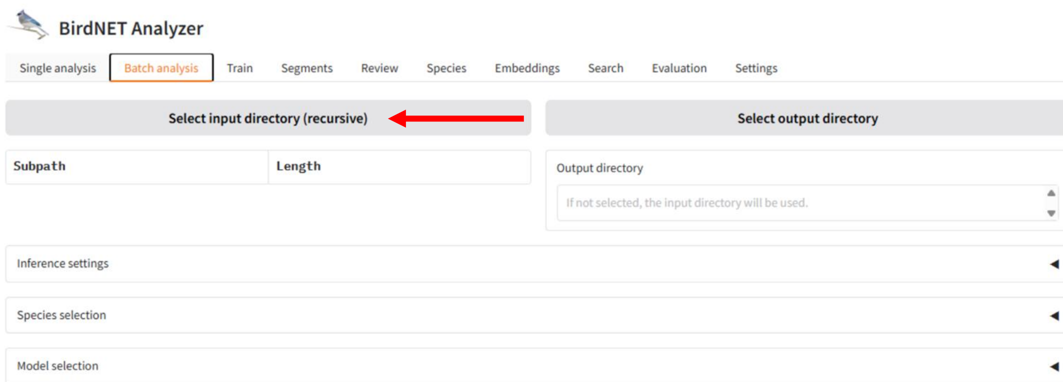
2.2 Running BirdNET Analyzer on Bird Data

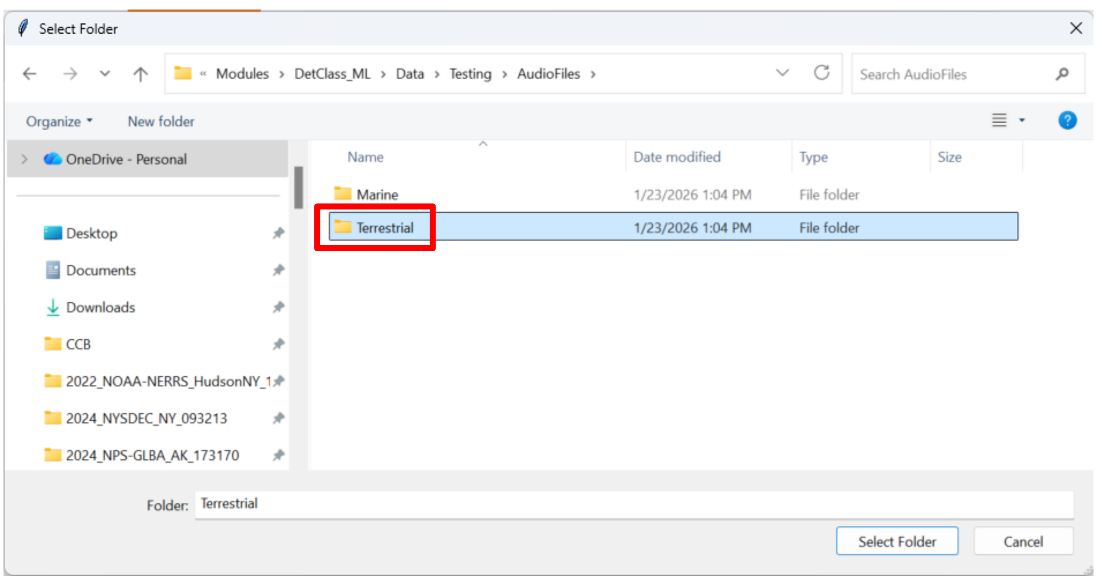
1. From your Desktop, **double-click the BirdNET icon to launch BirdNET-Analyzer**. It may take a minute for the program to open.

2. Navigate to the ***Batch analysis*** tab.

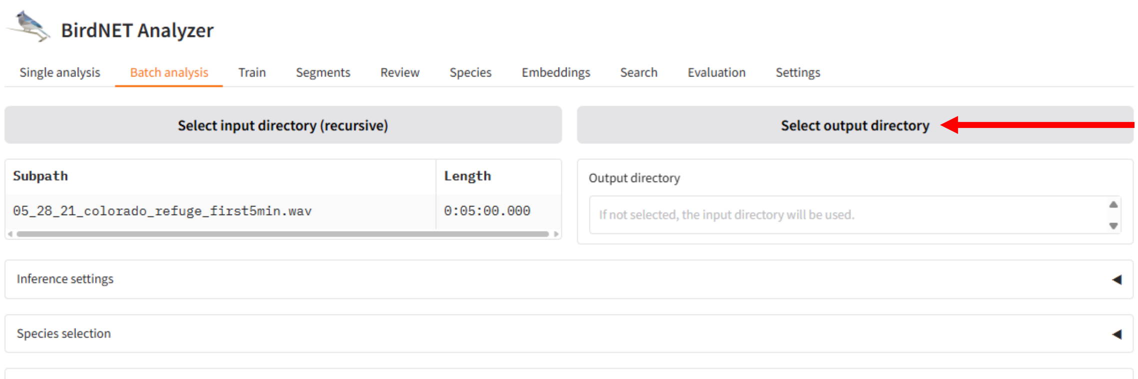


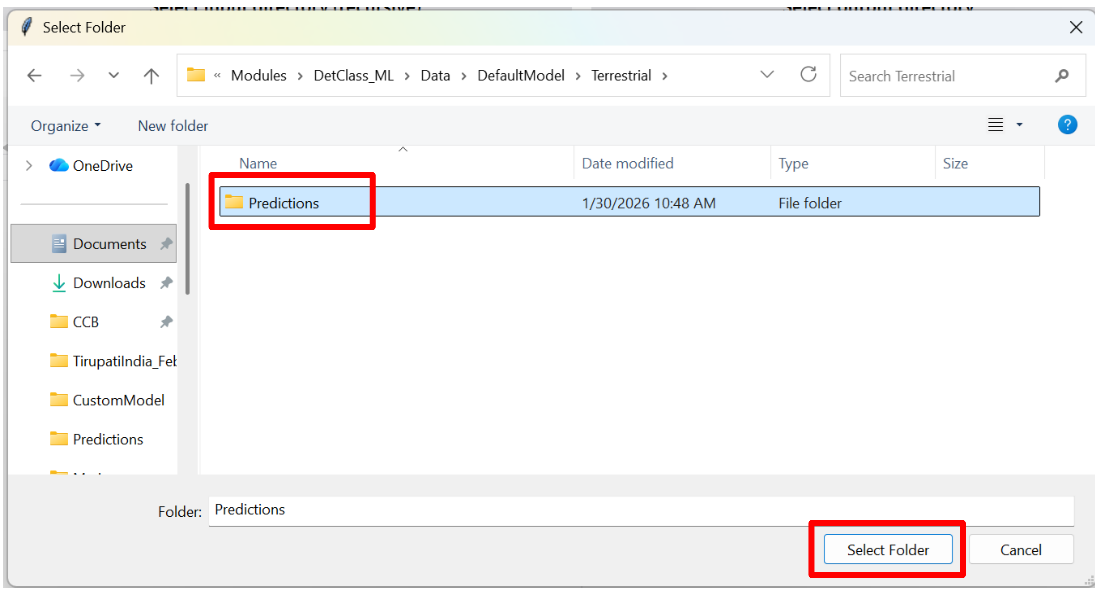
3. Click ***Select input directory*** and navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing >> Audio Files >> Terrestrial.*



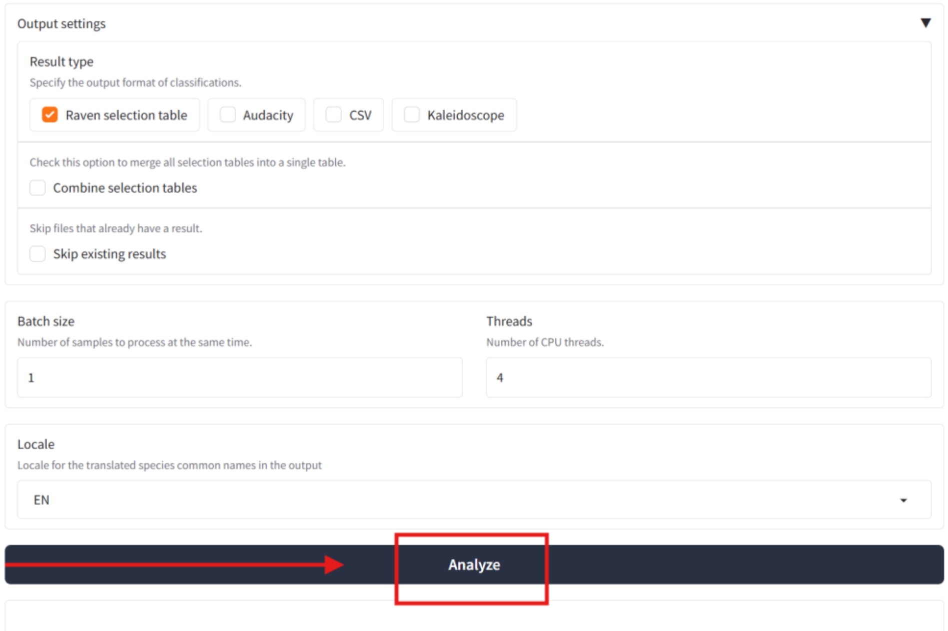


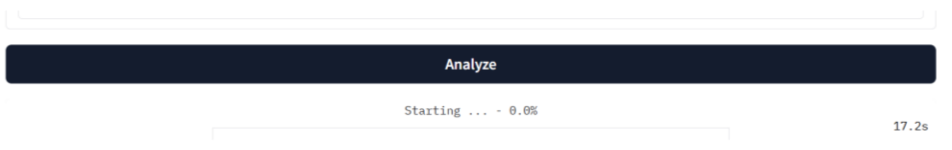
4. Click ***Select output directory and navigate to*** *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Terrestrial >> Predictions* .

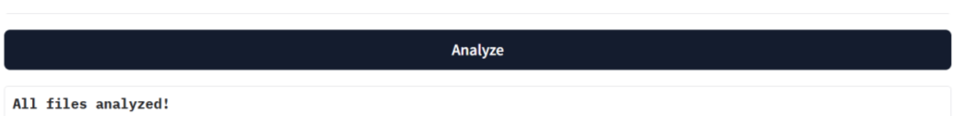




5. At the bottom of the screen, click ***Analyze.*** A progress bar will appear. When the analysis is complete it will switch to the message "All files analyzed!" Navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Terrestrial >> Predictions* to see the output selection table.

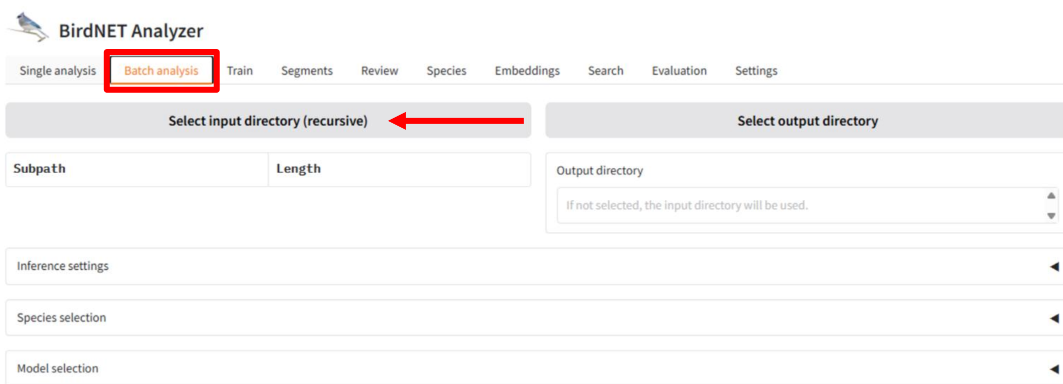

Output settings
Result type
Specify the output format of classifications.
Raven selection table
Audacity
CSV
Kaleidoscope
Check this option to merge all selection tables into a single table.
Combine selection tables
Skip files that already have a result.
Skip existing results
Batch size
Number of samples to process at the same time.
1
Threads
Number of CPU threads.
4
Locale
Locale for the translated species common names in the output
EN
Analyze

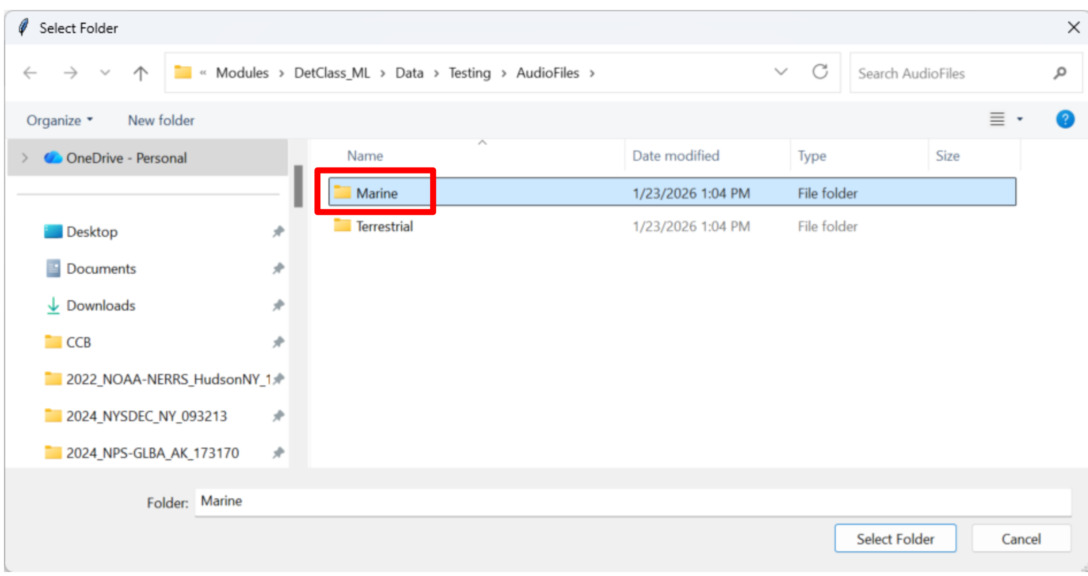

Analyze
Starting ... - 0.0%
17.2s


Analyze
All files analyzed!

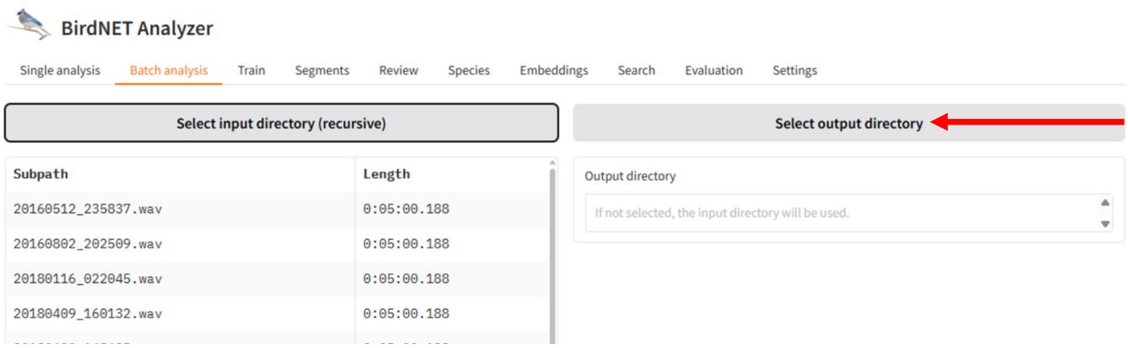
2.3 Running BirdNET Analyzer on Other Data

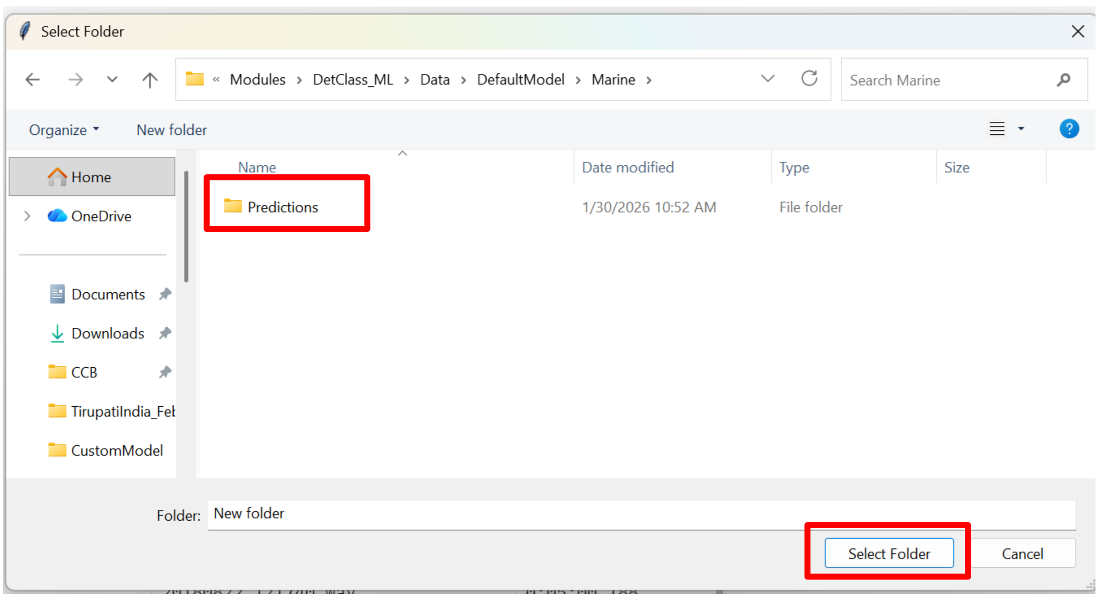
6. In the ***Batch analysis*** tab, click ***Select input directory*** and navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing >> Audio Files >> Marine.*





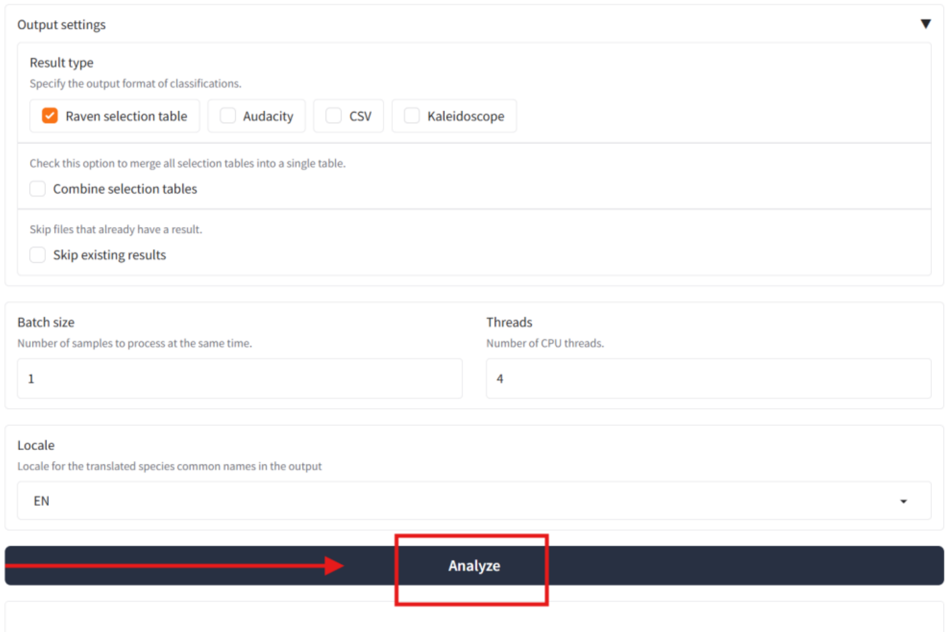
7. Click ***Select output directory*** and navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Marine >> Predictions* .

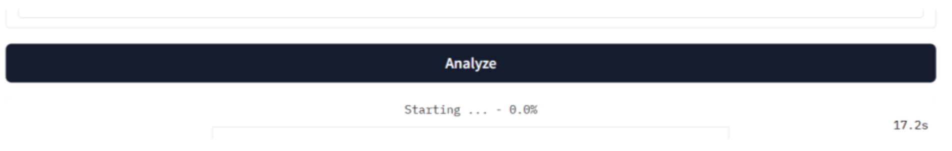


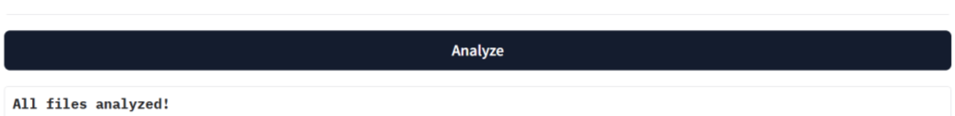


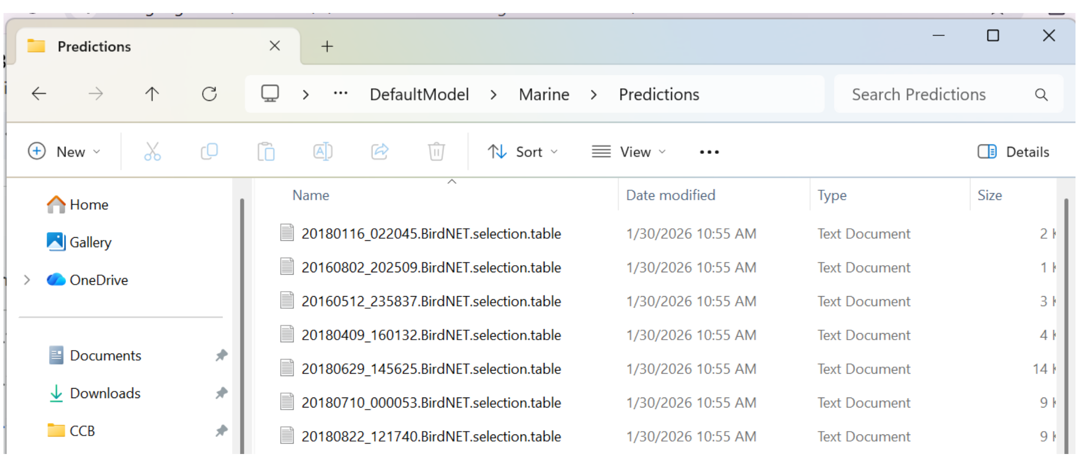
8. Under ***Inference settings***, set ***Maximum Bandpass Frequency (Hz)*** to 10000

9. At the bottom of the screen, click ***Analyze.*** A progress bar will appear. When the analysis is complete it will switch to the message "All files analyzed!" Navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Marine >> Predictions* to see the output selection tables.


Output settings
Result type
Specify the output format of classifications.
Raven selection table
Audacity
CSV
Kaleidoscope
Check this option to merge all selection tables into a single table.
Combine selection tables
Skip files that already have a result.
Skip existing results
Batch size
Number of samples to process at the same time.
1
Threads
Number of CPU threads.
4
Locale
Locale for the translated species common names in the output
EN
Analyze


Analyze
Starting ... - 0.0%
17.2s

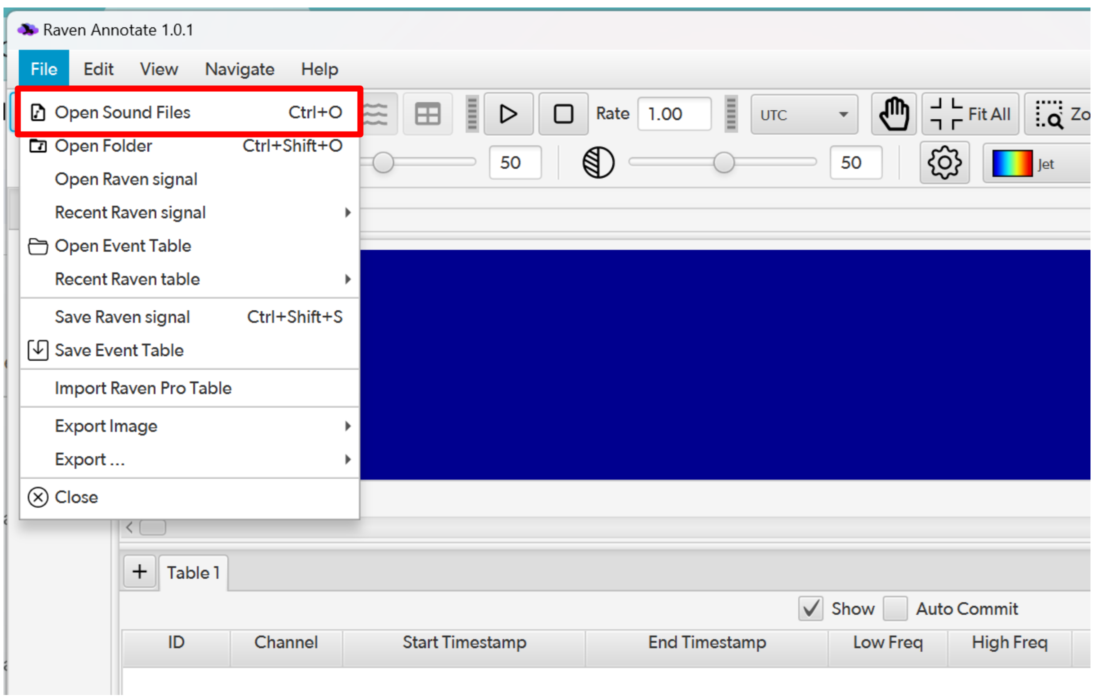

Analyze
All files analyzed!

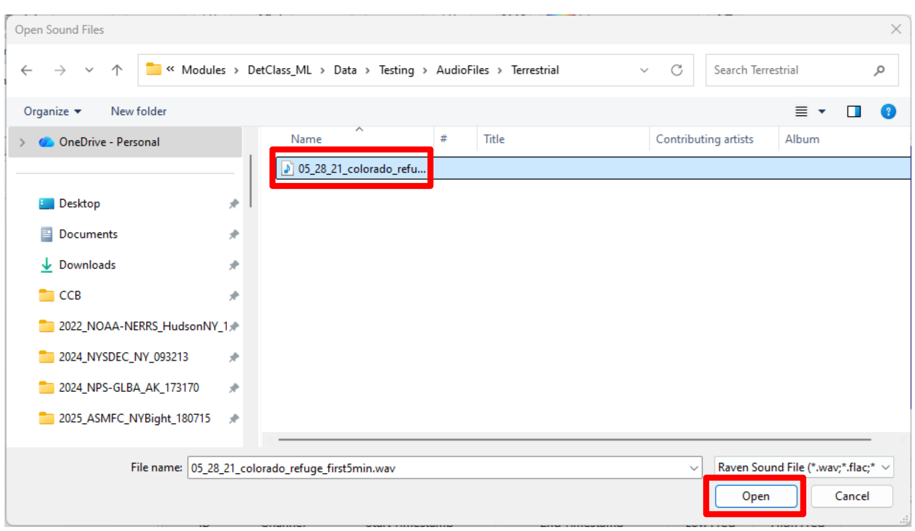

Predictions
DefaultModel
Marine
Predictions
Search Predictions
New
Sort
View
Details
Home
Gallery
OneDrive
Documents
Downloads
CCB
Name
Date modified
Type
Size
20180116_022045.BirdNET.selection.table
20160802_202509.BirdNET.selection.table
20160512_235837.BirdNET.selection.table
20180409_160132.BirdNET.selection.table
20180629_145625.BirdNET.selection.table
20180710_000053.BirdNET.selection.table
20180822_121740.BirdNET.selection.table
1/30/2026 10:55 AM
Text Document

Part 3: Performance Evaluation

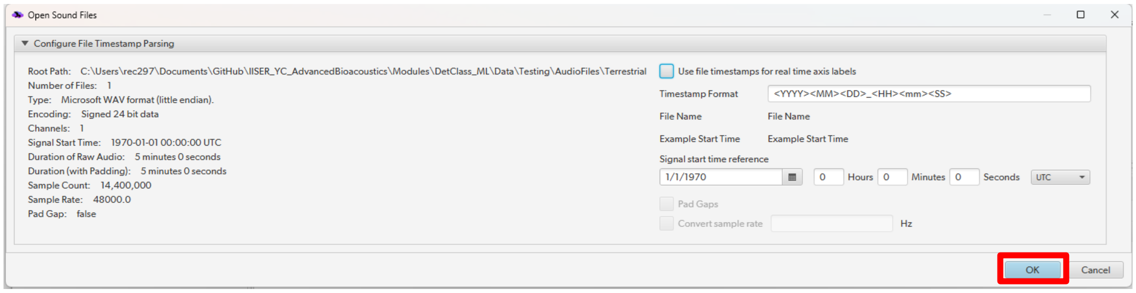
3.1 Viewing Annotations and Audio in Raven Annotate

1. Launch Raven Annotate, Pro, or Lite. Select ***`File >> Open Sound Files`***. Navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing >> Audio Files >> Terrestrial* and select the audio file 05_28_21_colorado_refuge_first5min.wav then click ***Open***.

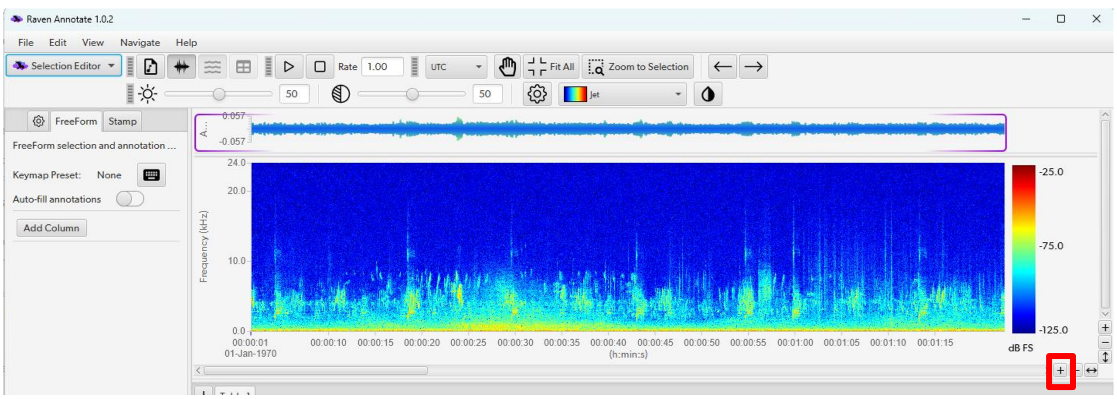




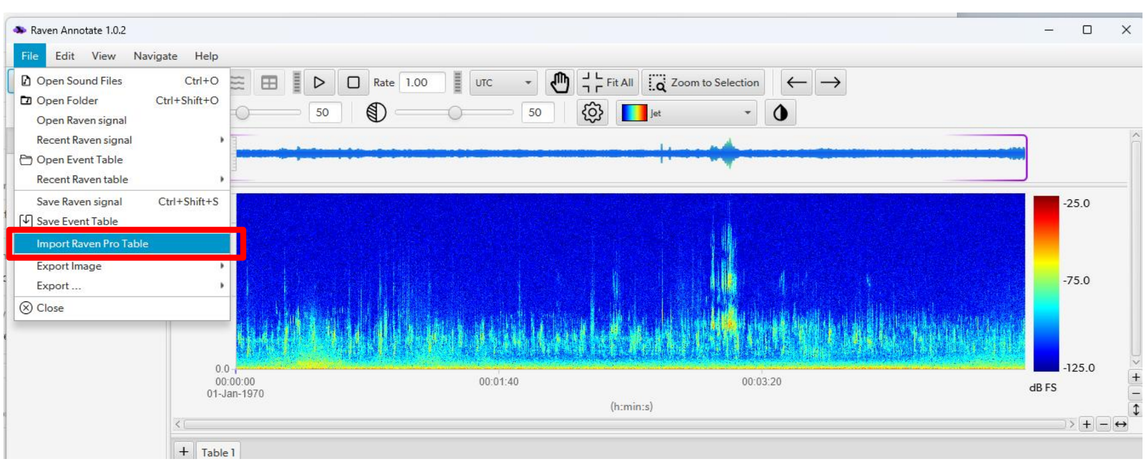
2. A dialogue box will appear. Uncheck the box for “Use file timestamps for real time axis labels”. Click ***OK*** and the data will load.

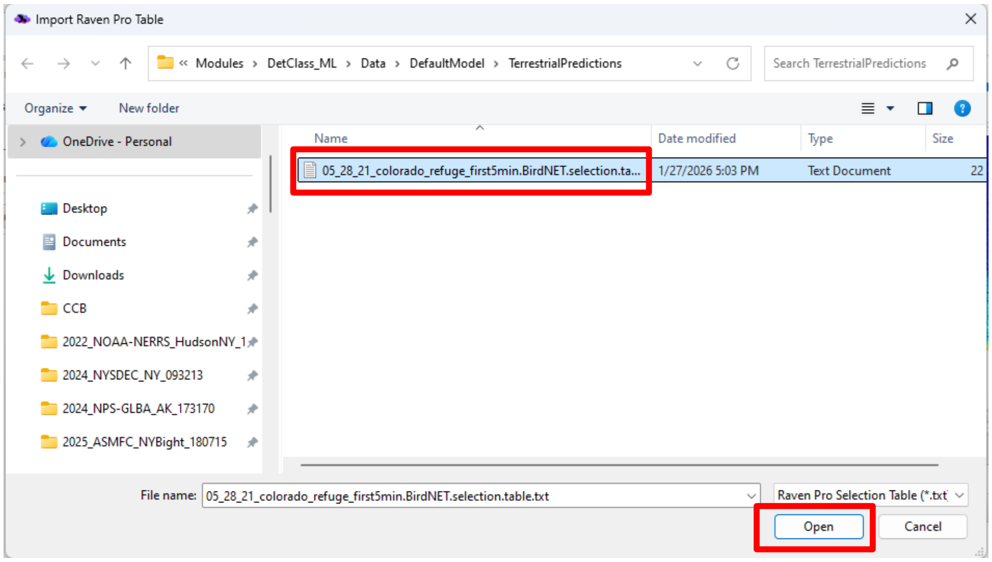


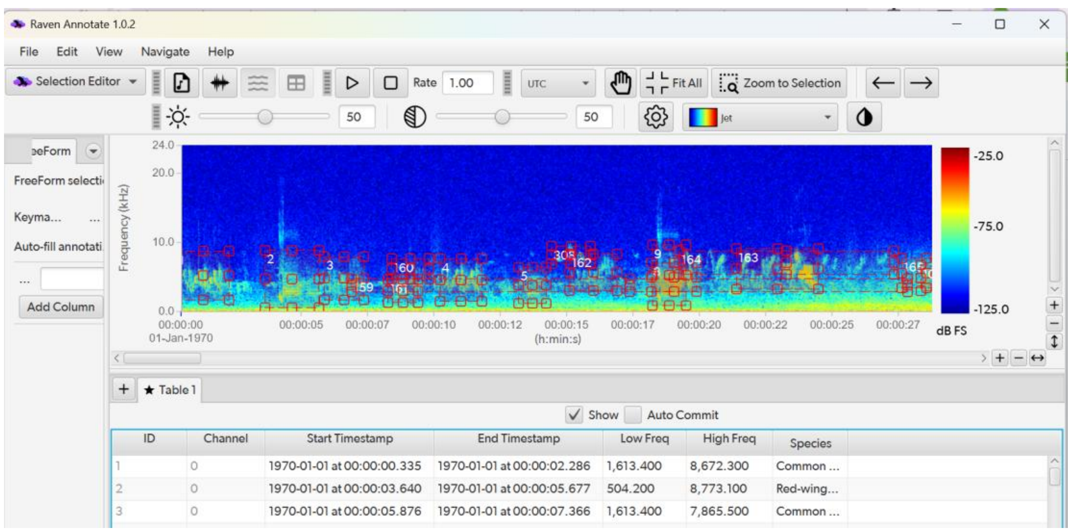
3. Zoom in on the x-axis by clicking the **+** in the bottom right corner of the spectrogram panel. Use the x-axis scroll bar to navigate to the start of the file.



4. Select ***File >> Import Raven Pro Table***. Navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel>> Terrestrial >> Predictions* and select 5_28_21_colorado_refuge_first5min.BirdNET.selection.table.txt, then click ***Open***.

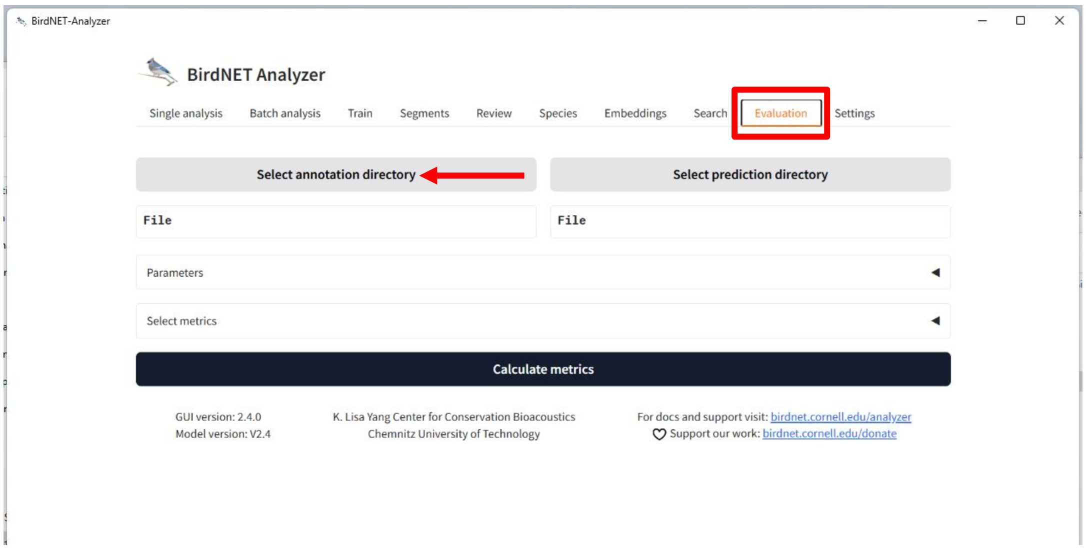


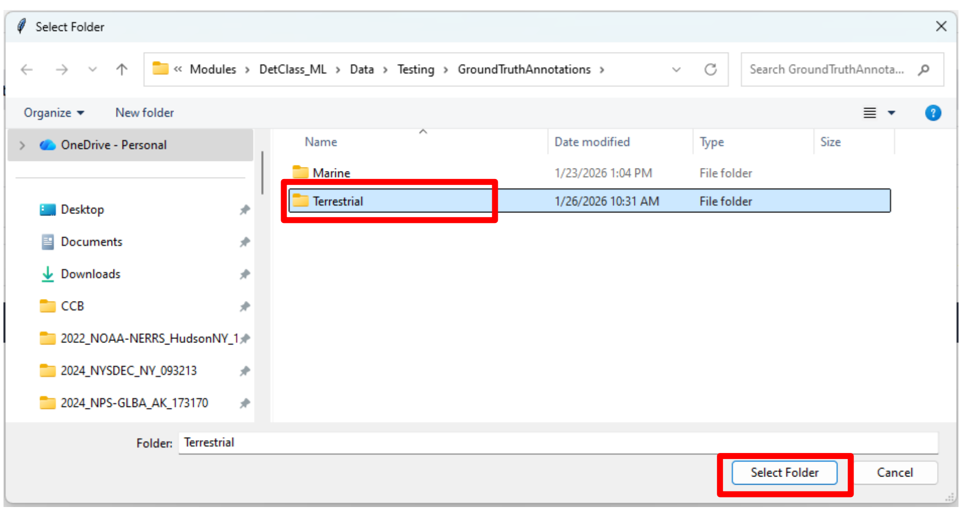

Import Raven Pro Table
Modules › DetClass_ML › Data › DefaultModel › TerrestrialPredictions
Search TerrestrialPredictions
Organize
New folder
OneDrive - Personal
Desktop
Documents
Downloads
CCB
2022_NOAA-NERRS_HudsonNY_1
2024_NYSDEC_NY_093213
2024_NPS-GLBA_AK_173170
2025_ASMFC_NYBight_180715
Name
Date modified
Type
Size
05_28_21_colorado_refuge_first5min.BirdNET.selection.ta...
1/27/2026 5:03 PM
Text Document
File name:
05_28_21_colorado_refuge_first5min.BirdNET.selection.table.txt
Raven Pro Selection Table (*.txt
Open
Cancel


Raven Annotate 1.0.2
File
Edit
View
Navigate
Help
Selection Editor
Rate
1.00
UTC
Fit All
Zoom to Selection
Frequency (kHz)
(h:min:s)
dB FS
Add Column
Table 1
Show
Auto Commit
ID
Channel
Start Timestamp
End Timestamp
Low Freq
High Freq
Species
1 0 1970-01-01 at 00:00:00.335 1970-01-01 at 00:00:02.286 1,613.400 8,672.300 Common ...
2 0 1970-01-01 at 00:00:03.640 1970-01-01 at 00:00:05.677 504.200 8,773.100 Red-wing...
3 0 1970-01-01 at 00:00:05.876 1970-01-01 at 00:00:07.366 1,613.400 7,865.500 Common ...

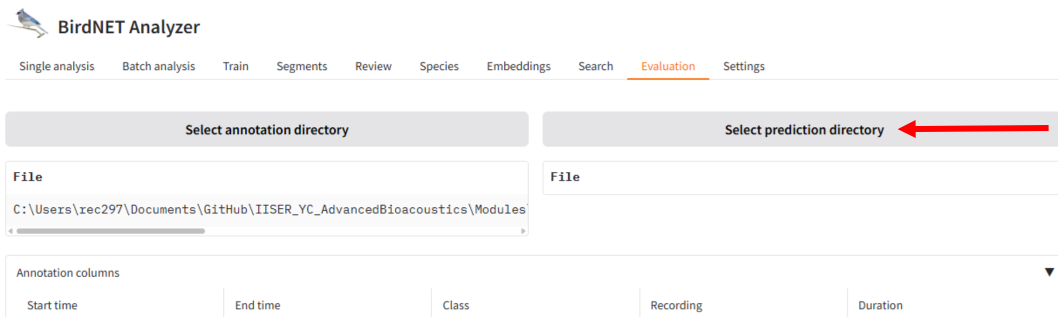
3.2 Quantifying Model Performance with BirdNET

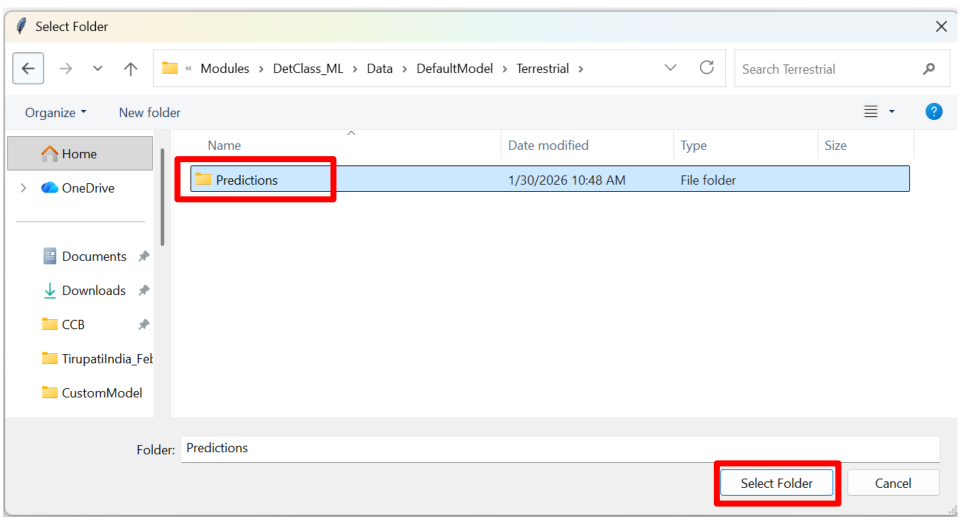
1. Click on the ***Evaluation*** tab. Click ***Select annotation directory,*** navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing >> GroundTruthAnnotations>>Terrestrial*, and click ***Select Folder***.



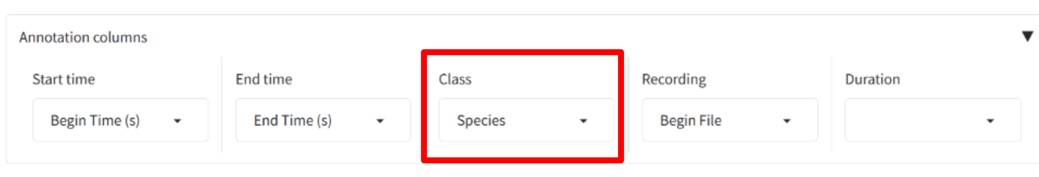


2. Click ***Select prediction directory,*** navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Terrestrial >> Predictions*, and click ***Select***

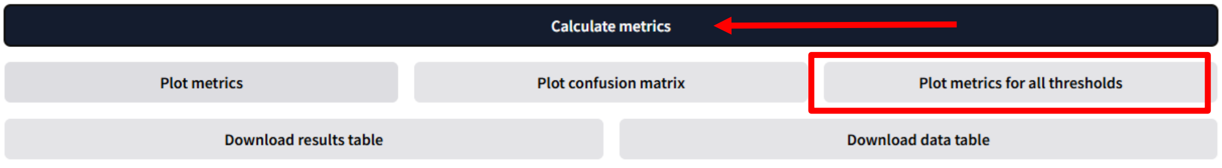




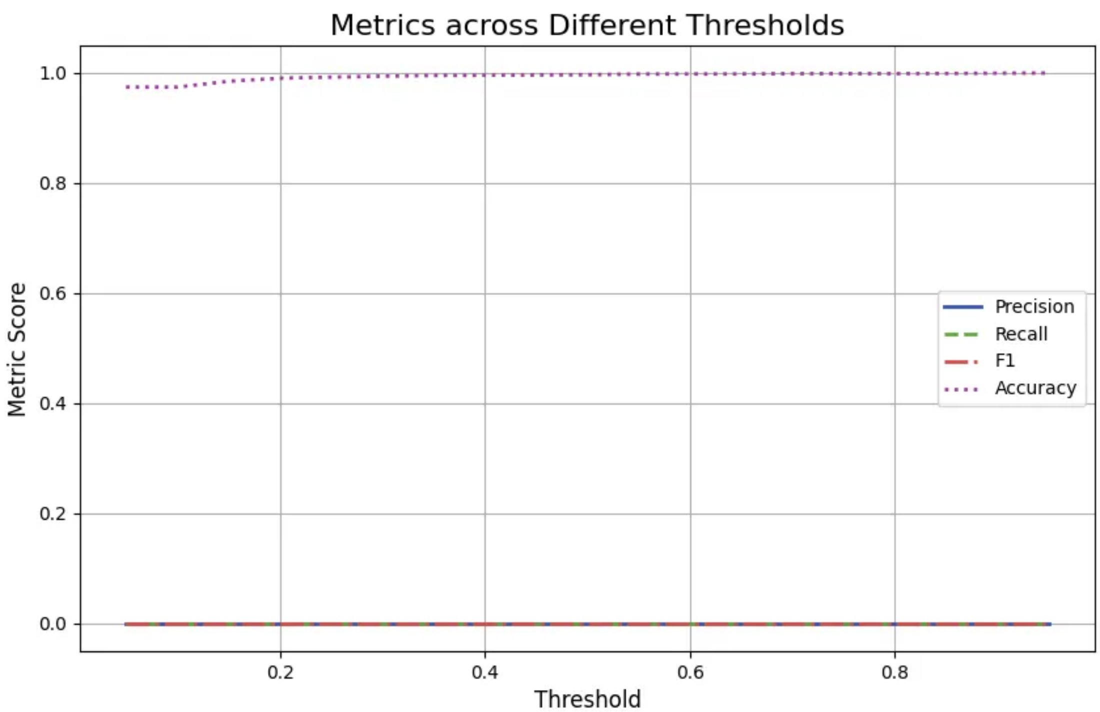
3. Scroll down to ***Annotation columns***. Under ***Class***, use the drop down menu to select ***Species***.



4. Click ***Calculate metrics***. A results table with precision and recall will be generated. Plot precision/recall/F1/accuracy vs. threshold by clicking ***Plot metrics for all thresholds***.



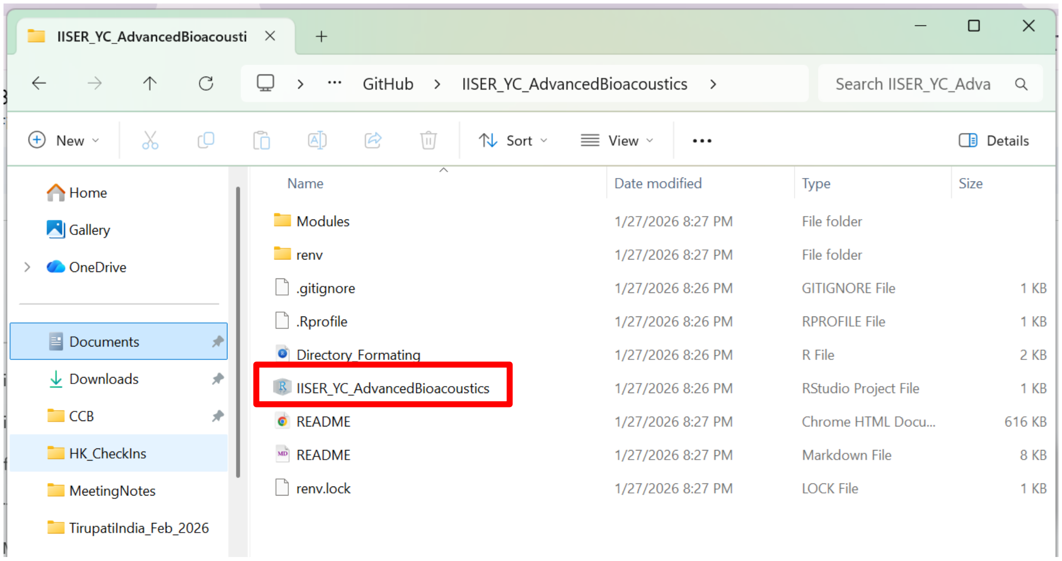
	AUROC	Precision	Recall	F1	AP	Accuracy
Overall	null	0	0	0	0	0.9736111111111112



5. Repeat process with marine ground truth (*IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing >> GroundTruthAnnotations>>Marine*) and marine BirdNET predictions (*IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Marine >> Predictions*).

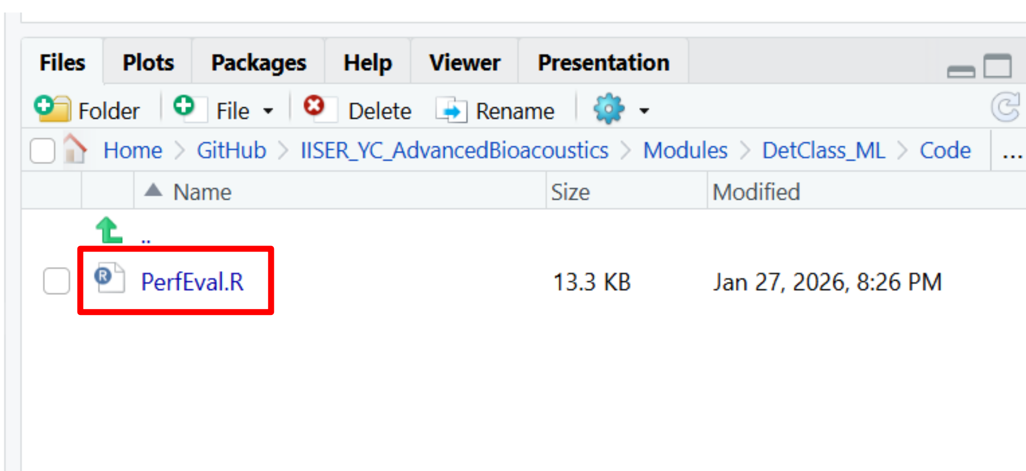
3.3 Quantifying Model Performance with Custom Scripting

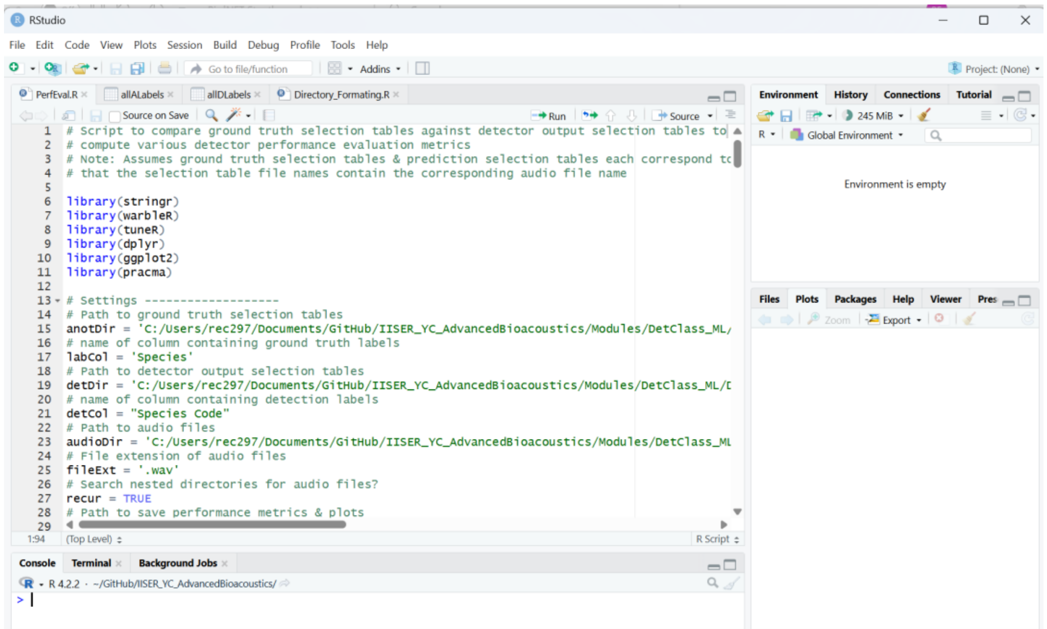
1. Launch the workshop R Project by navigating to the downloaded GitHub repository folder *IISER_YC_AdvancedBioacostics* and double clicking the file `IISER_YC_AdvancedBioacoustics.Rproj`.



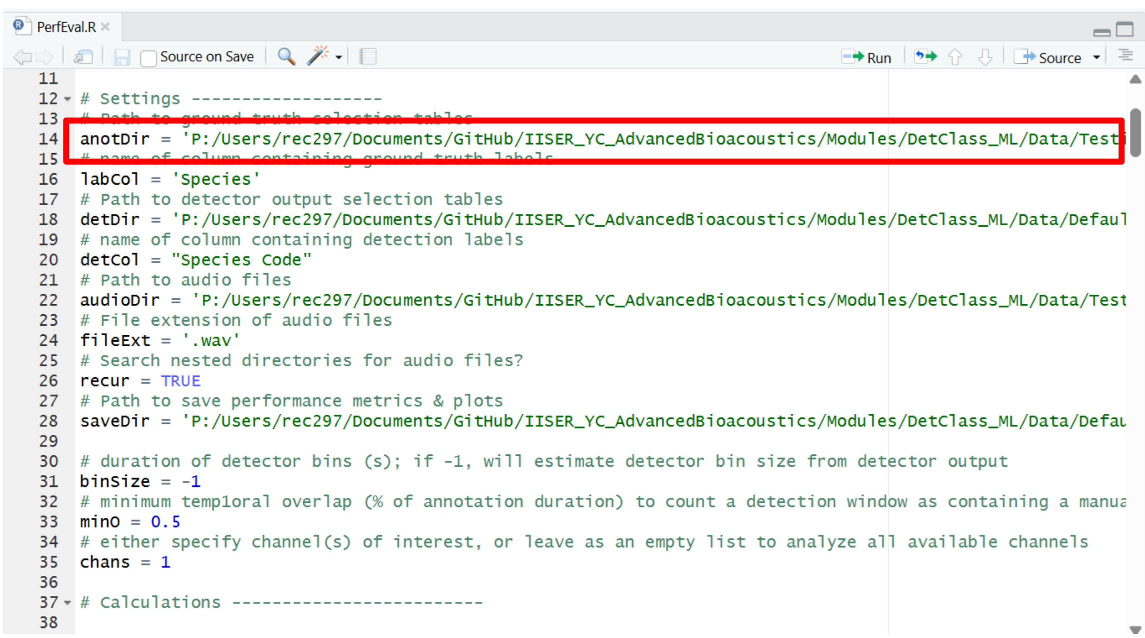
2. In the ***Console*** enter the command `renv::restore()`

3. In the ***Files*** pane of the RStudio session, navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Code* and click `PerfEval.R` to open the script.

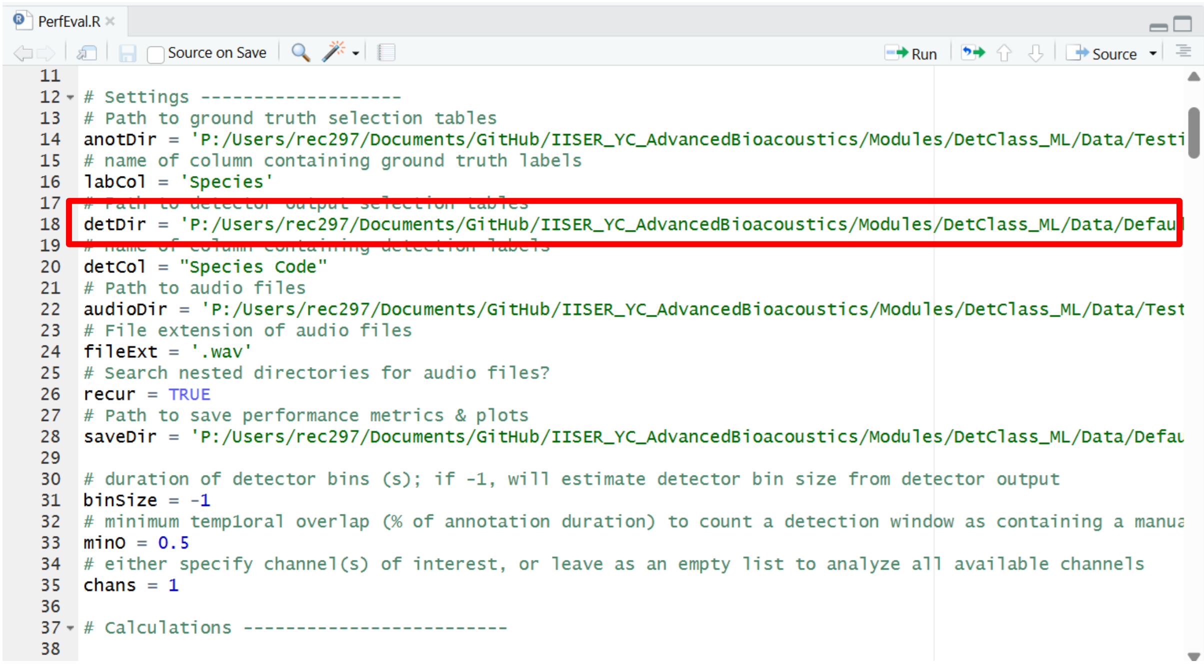




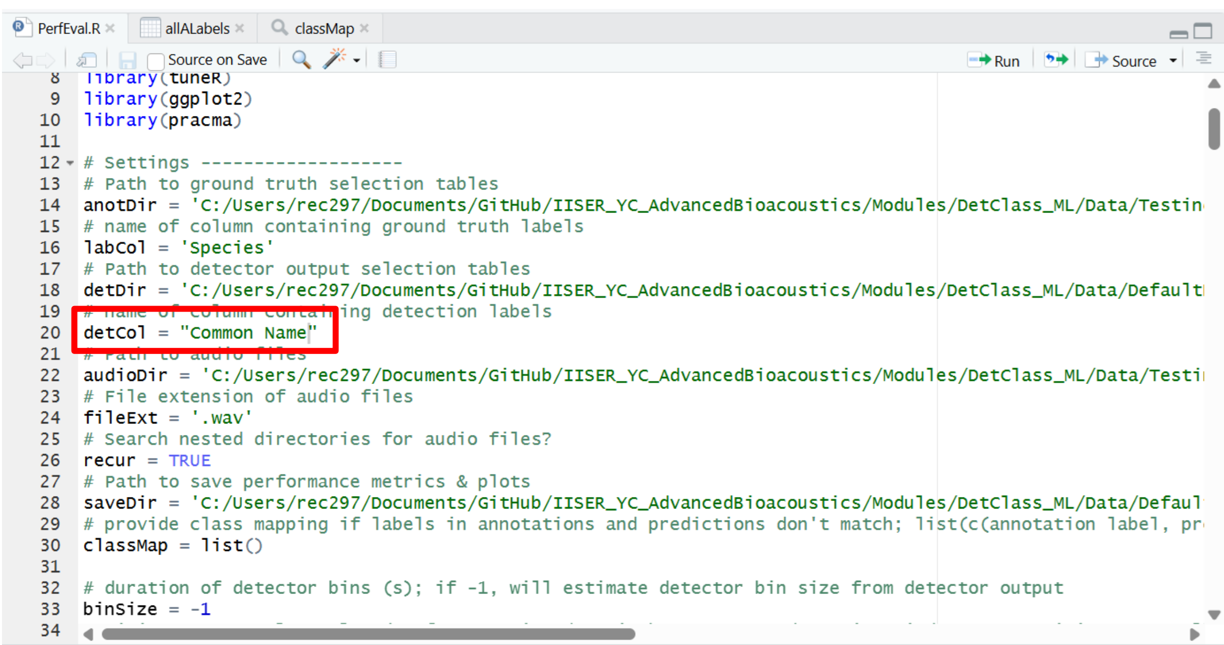
4. Update the `anotDir` variable path in line 14 to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing>> GroundTruthAnnotations >> Terrestrial*



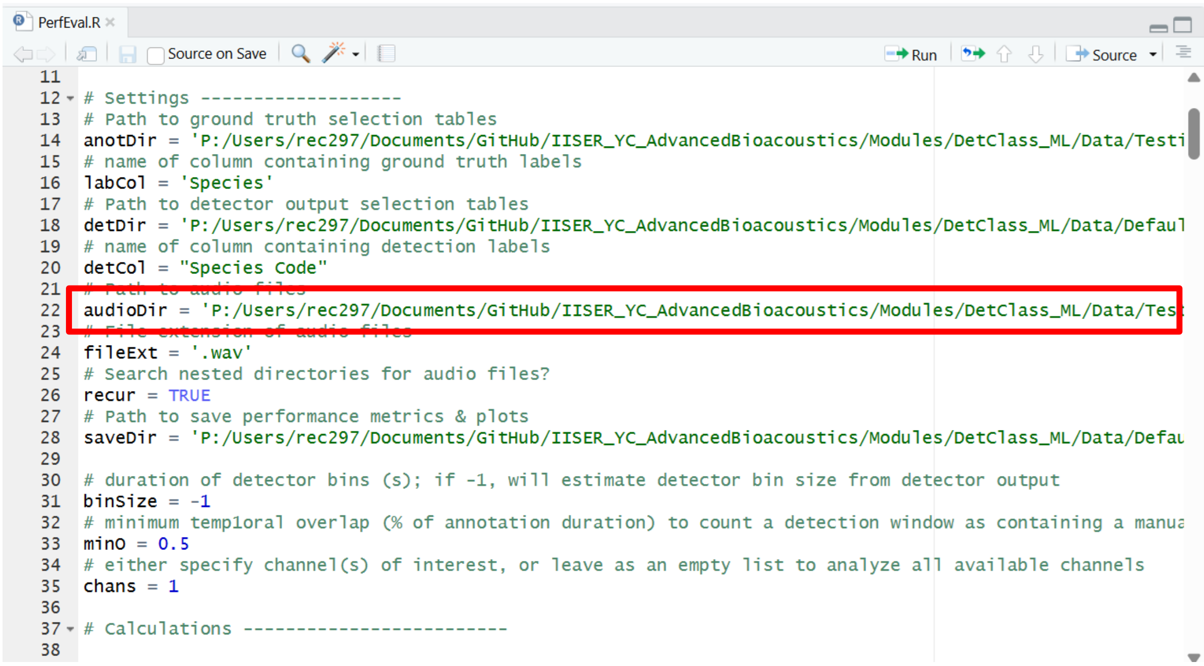
5. Update the `detDir` variable path in line 18 to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Terrestrial >> Predictions*



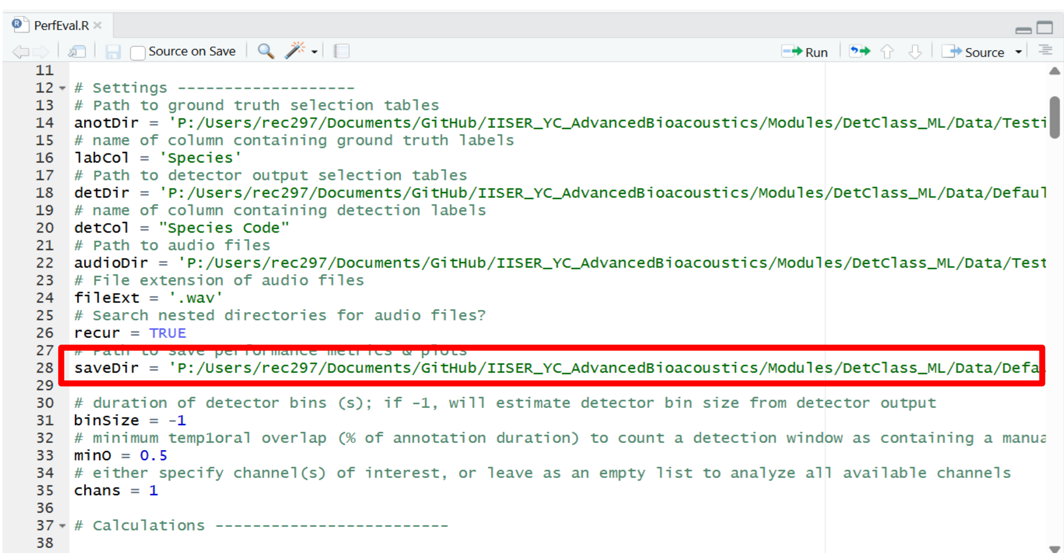
6. Update the `detCol` variable in line 20 to “Common Name”



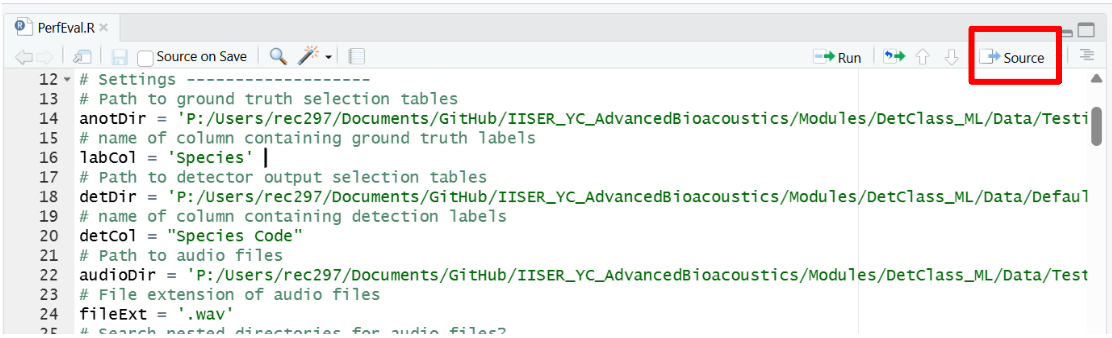
7. Update the `audioDir` variable path in line 22 to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing>> AudioFiles >> Terrestrial*



8. Update the `saveDir` variable path in line 28 to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Terrestrial*



9. Run script by clicking ***Source*** button in upper right corner of script editor window



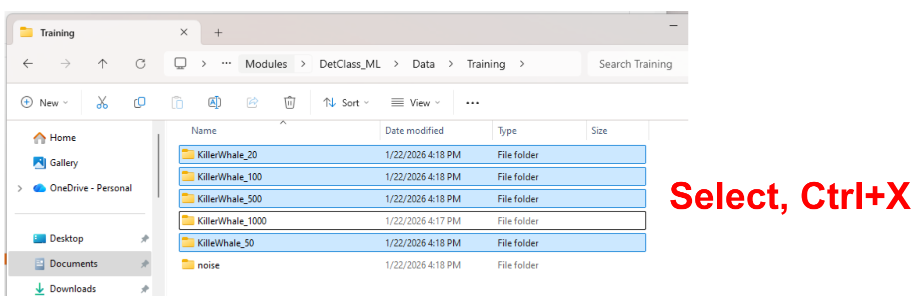
10. In Windows Explorer or FInder, navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Terrestria >> lPredictions* to view output figures.
11. Repeat process for marine data predictions.

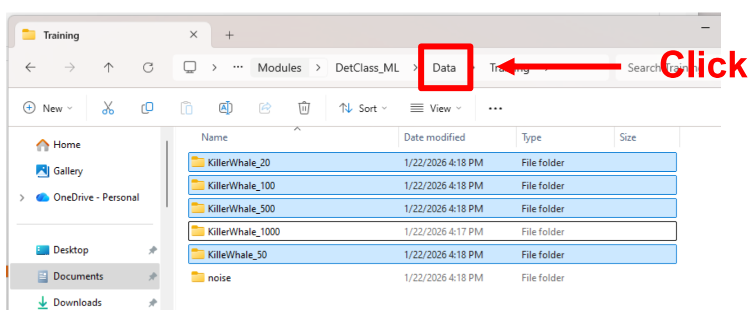
Part 4: Custom BirdNET Models

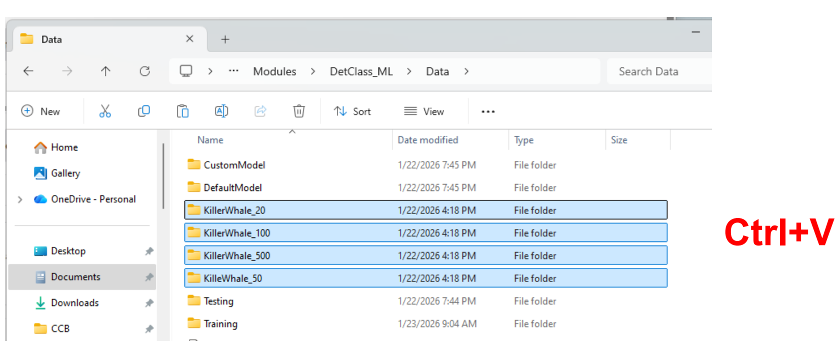
We can use BirdNET to detect sound classes that it was not originally trained on by applying transfer learning with a modest amount of training data. To train a custom BirdNET model, you need a training dataset consisting of short sound clips from both the target class(es) (specific species or sound type(s)) and at least one non-target class (e.g. background noise).

4.1 Training a Custom Model

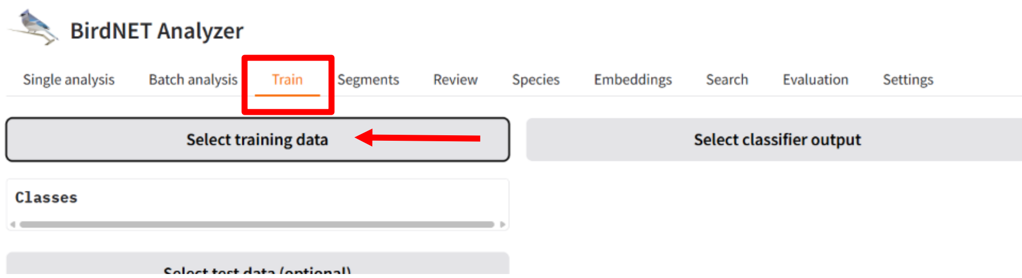
1. We will train a custom model with two classes: Killer Whale and Noise. BirdNET will use all training data folders in the directory you point it towards, so we'll need to reorganize the data slightly so that each team uses a different number of training examples. In the workshop folder, navigate to *Modules >> DetClass_ML >> Data >> Training*. **Select and cut all killer whale training data folders *except the one assigned to your group*. Paste them up one level, directly into the Data folder.**

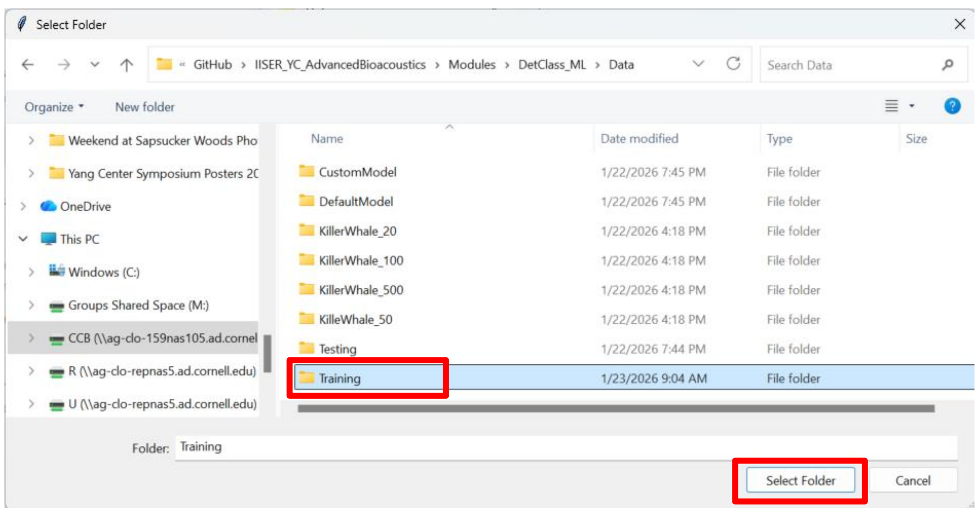




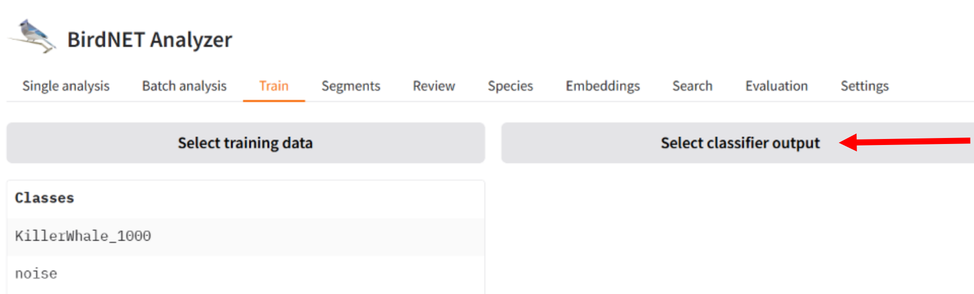


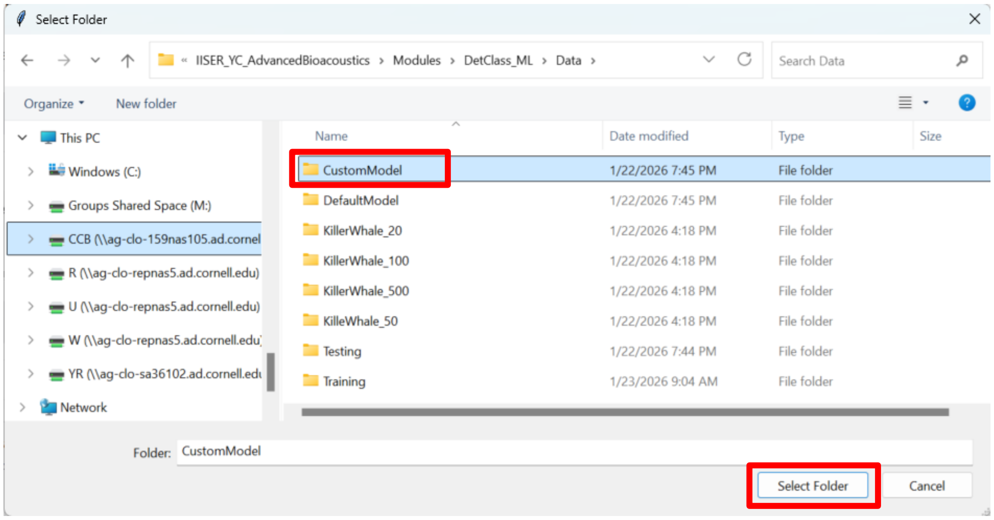
2. In ***BirdNET Analyzer***, navigate to the ***Train*** tab. Click ***Select training data***, navigate to *Modules >> DetClass_ML >> Data >> Training*, and click ***Select***.

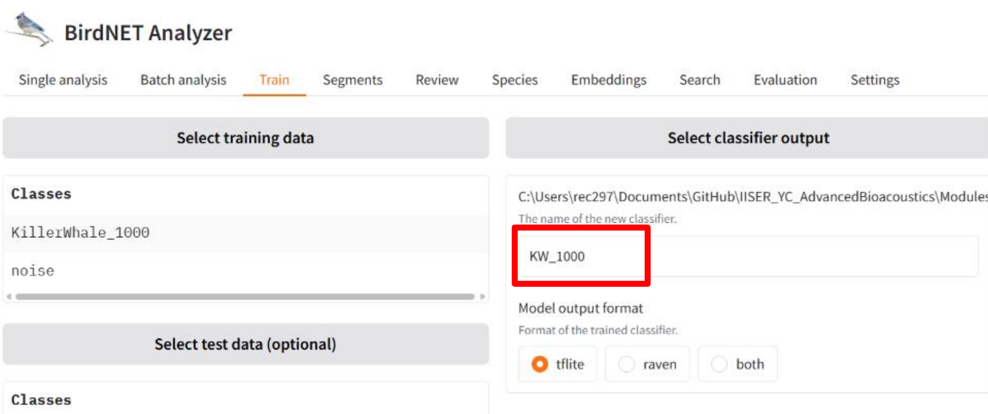




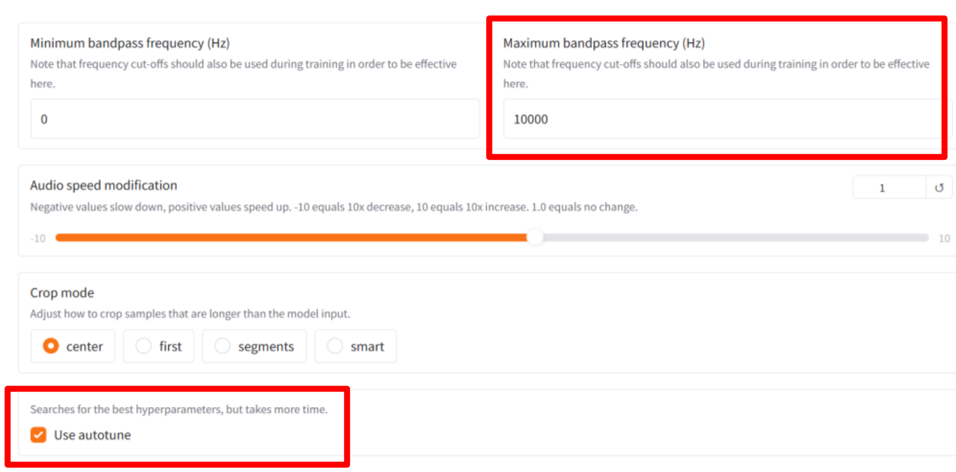
3. Click ***Select classifier output***, navigate to *Modules >> DetClass_ML >> Data >> CustomModel*, and click ***Select***. You can also specify a name for your custom model (by default it will be called "CustomClassifier").



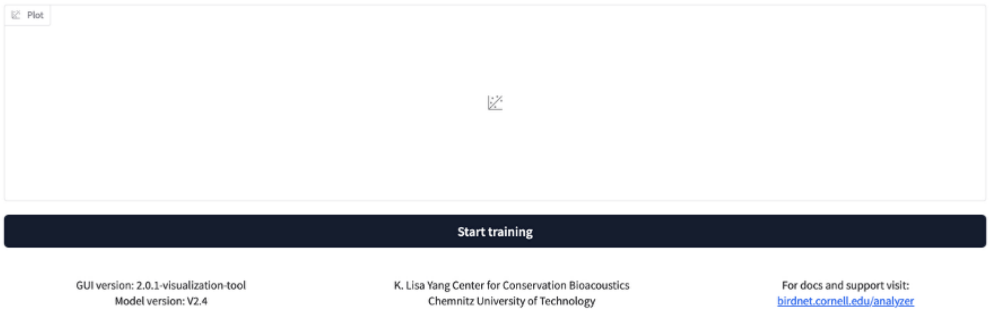




4. Set the ***Maximum bandpass frequency (Hz)*** to 10000. Click the box next to ***Use autotune*** to implement automatic hyperparameter optimization.

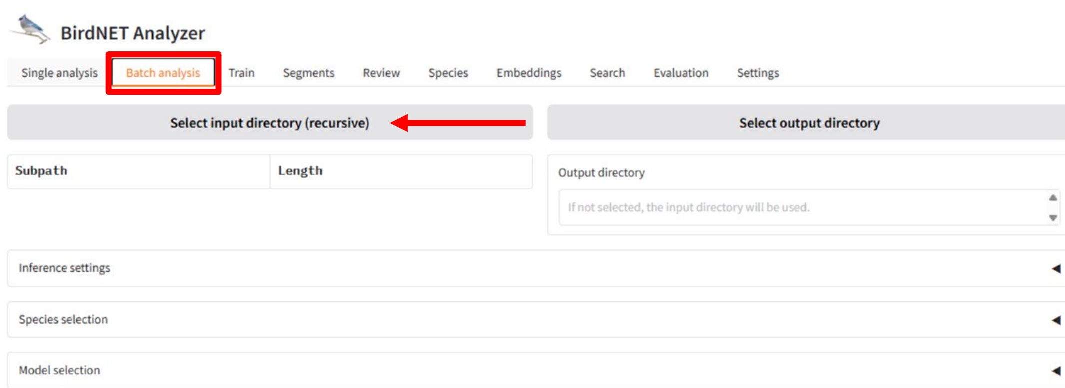


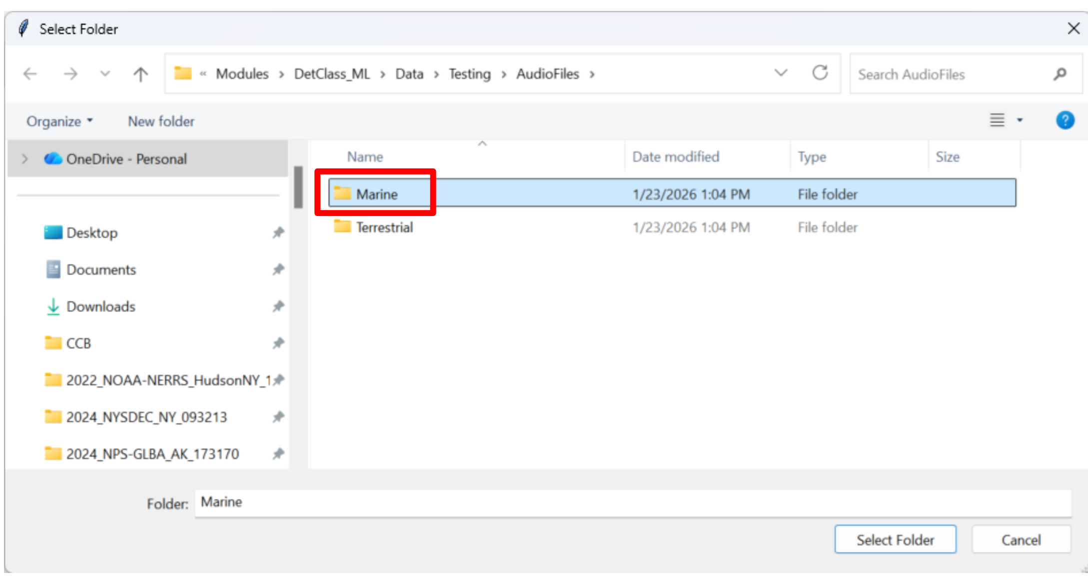
5. Scroll to the bottom of the train tab and click ***Start training.***



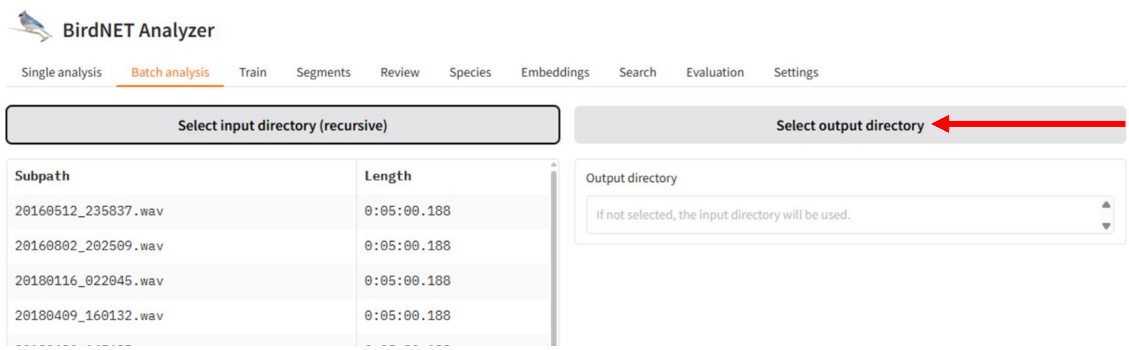
4.2 Deploying a Custom Model

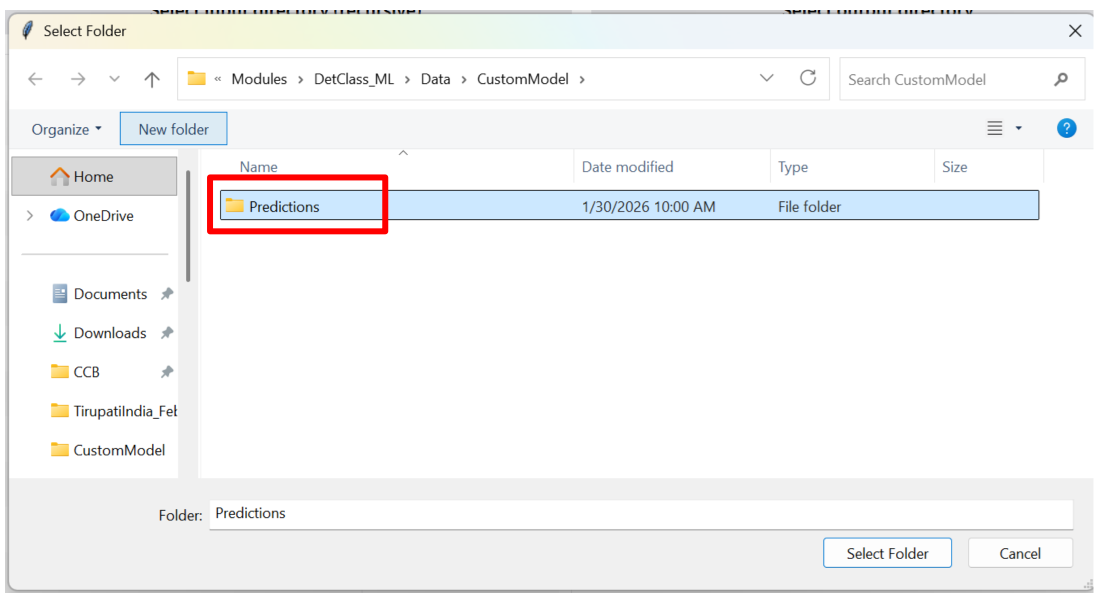
1. In the ***Batch analysis*** tab, click ***Select input directory*** and navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing >> Audio Files >> Marine.*



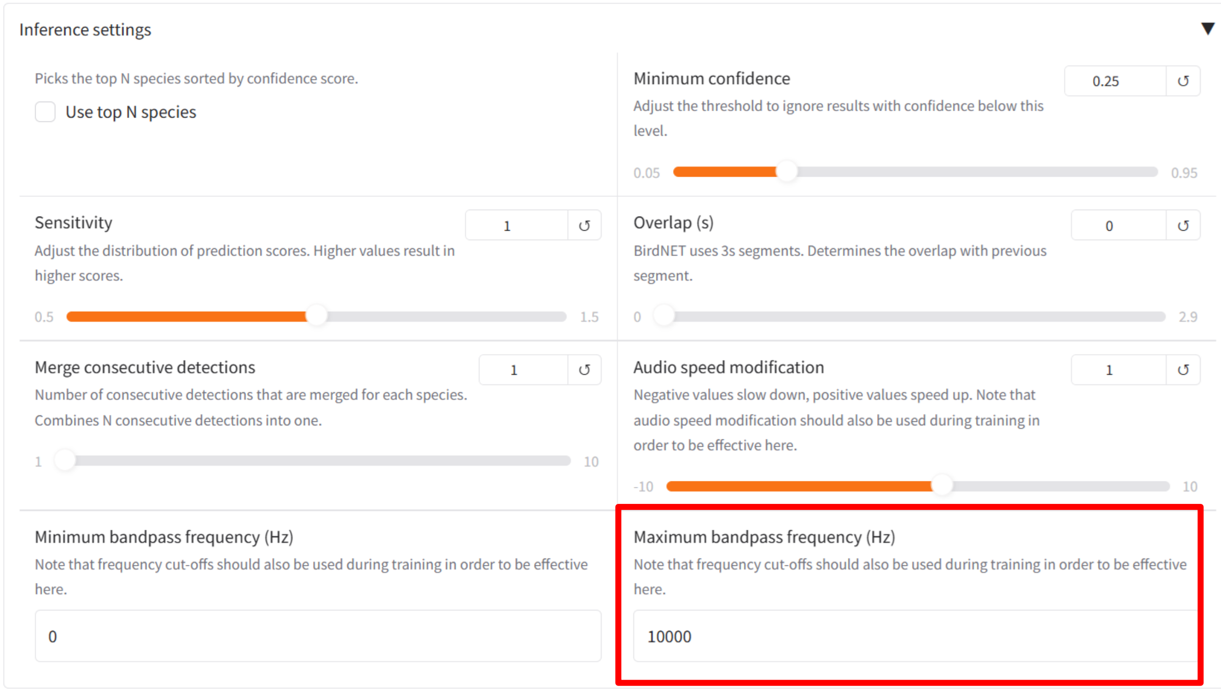


2. Click ***Select output directory and navigate to*** *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> CustomModel*, select *Predictions*, and click ***Select folder***.

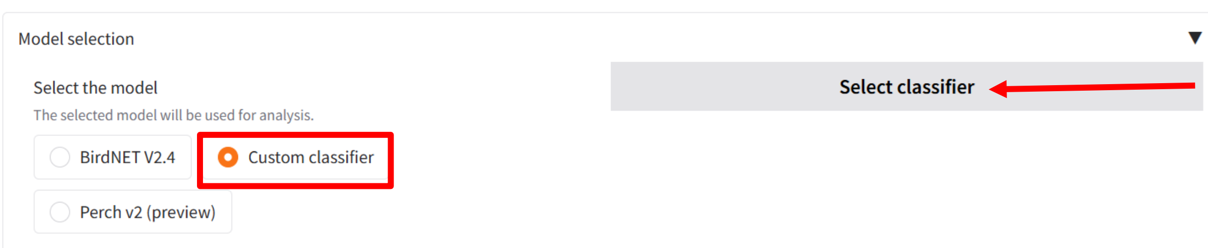


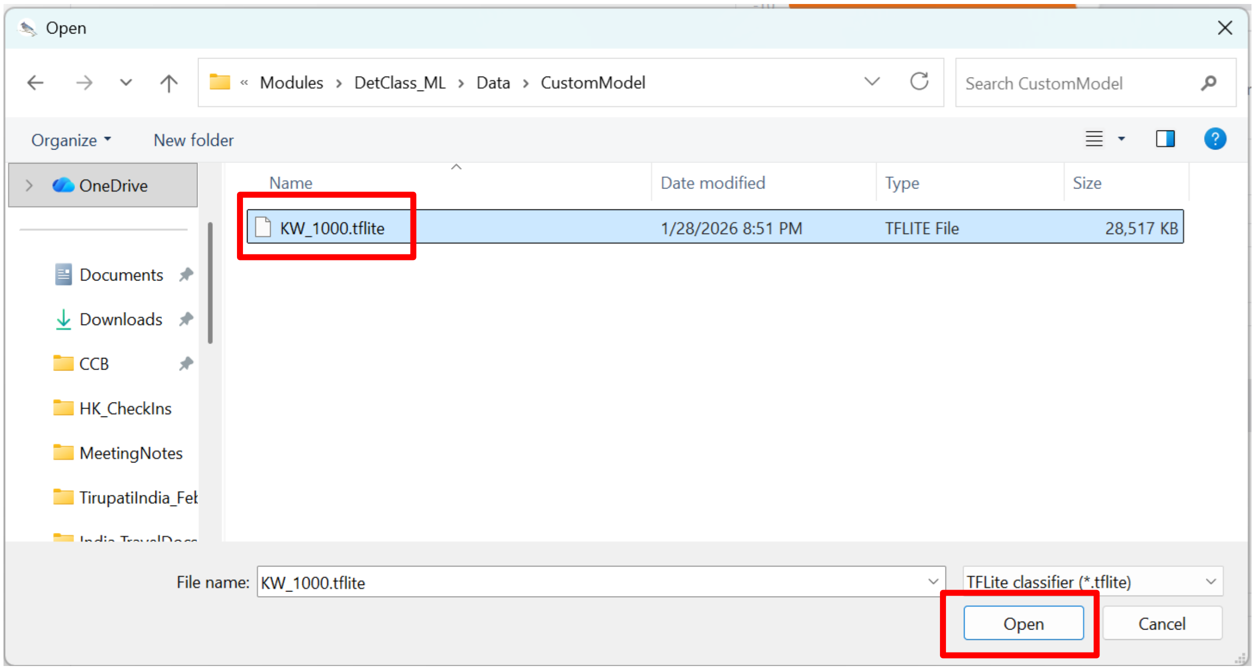


3. Under ***Inference settings***, set ***Maximum Bandpass Frequency (Hz)*** to 10000

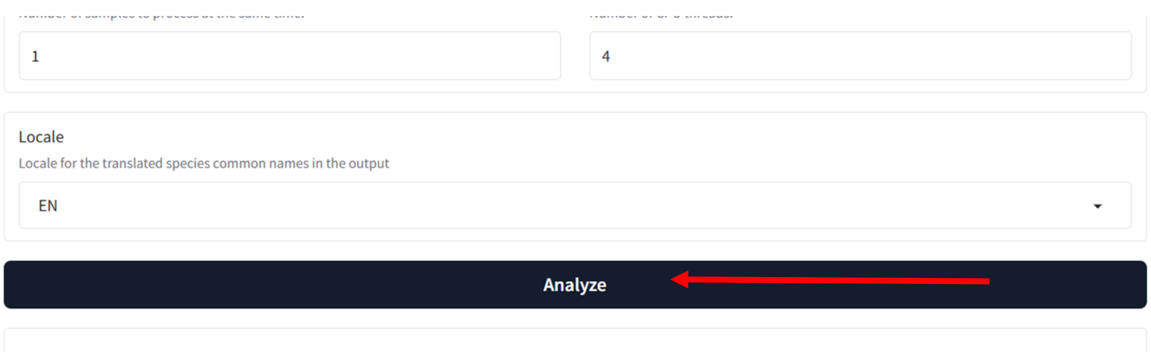


4. Under ***Model selection***, select ***Custom classifier***, then click the ***Select classifier*** button and navigate to *Modules >> DetClass_ML >> Data >> CustomModel*, select the classifier you just trained and click ***Open***.





5. Scroll to the bottom of the window and click ***Analyze*** to deploy the custom model.



4.3 Evaluating Custom Model Performance

1. Update the `anotDir` variable path in line 14 to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing>> GroundTruthAnnotations >> Marine*

PerfEval.R

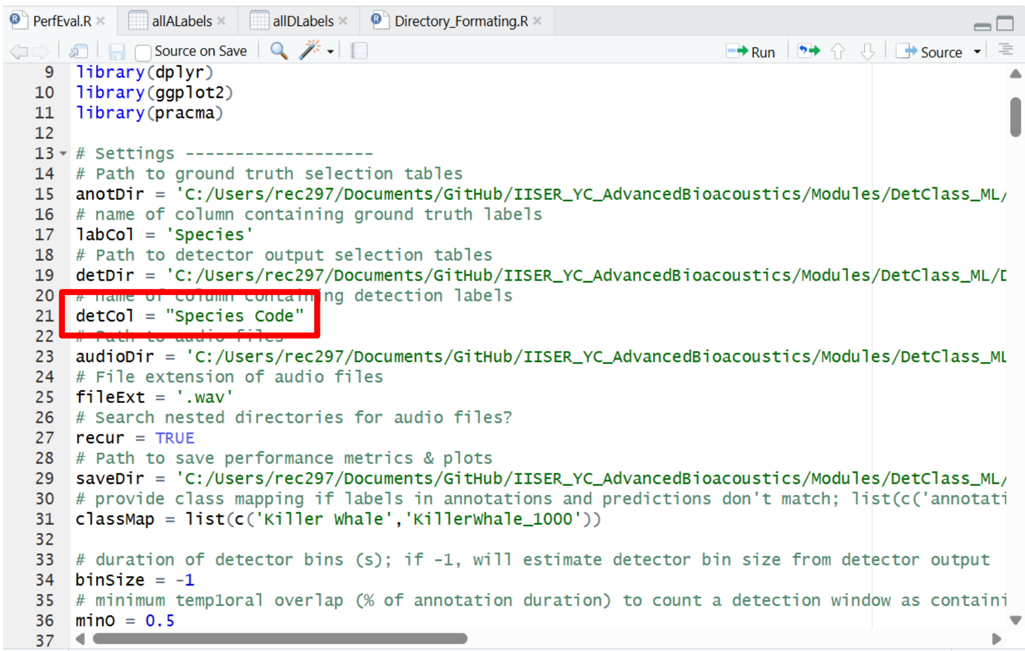
```
11
12 # Settings -------------------
13 # Path to ground truth selection tables
14 anotDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Test
15 # name of column containing ground truth labels
16 labCol = 'Species'
17 # Path to detector output selection tables
18 detDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Defaul
19 # name of column containing detection labels
20 detCol = "Species Code"
21 # Path to audio files
22 audioDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Test
23 # File extension of audio files
24 fileExt = '.wav'
25 # Search nested directories for audio files?
26 recur = TRUE
27 # Path to save performance metrics & plots
28 saveDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Defau
29
30 # duration of detector bins (s); if -1, will estimate detector bin size from detector output
31 binSize = -1
32 # minimum temploral overlap (% of annotation duration) to count a detection window as containing a manua
33 minO = 0.5
34 # either specify channel(s) of interest, or leave as an empty list to analyze all available channels
35 chans = 1
36
37 # Calculations -------------------------
38
```

2. Update the `detDir` variable path in line 18 to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> CustomModel >> Predictions*

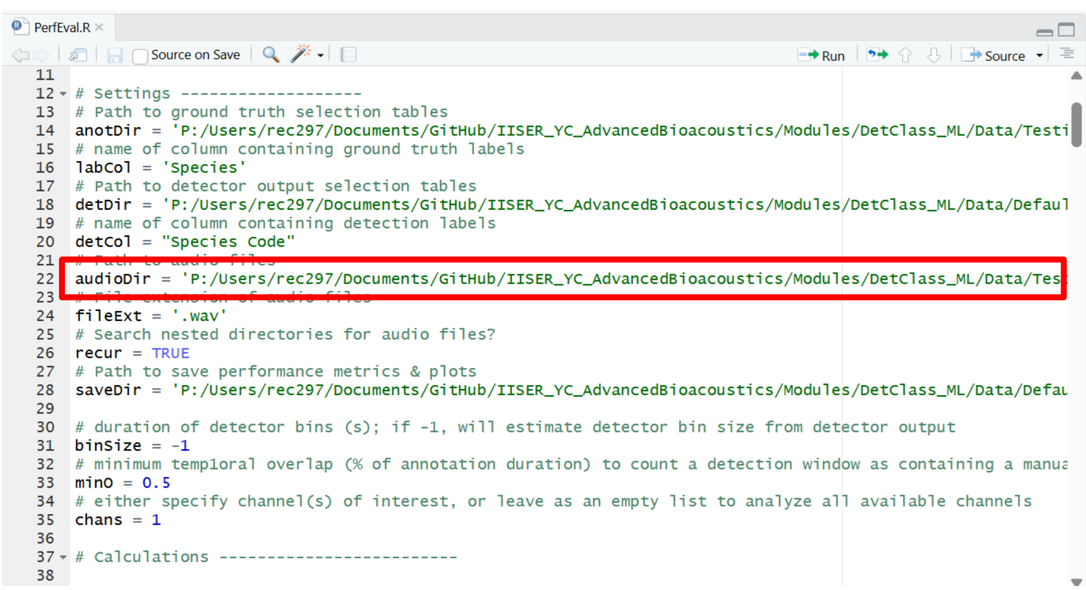
PerfEval.R

```
11
12 # Settings -------------------
13 # Path to ground truth selection tables
14 anotDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testi
15 # name of column containing ground truth labels
16 labCol = 'Species'
17 # Path to detector output selection tables
18 detDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Defau
19 # name of column containing detection labels
20 detCol = "Species Code"
21 # Path to audio files
22 audioDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Test
23 # File extension of audio files
24 fileExt = '.wav'
25 # Search nested directories for audio files?
26 recur = TRUE
27 # Path to save performance metrics & plots
28 saveDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Defau
29
30 # duration of detector bins (s); if -1, will estimate detector bin size from detector output
31 binSize = -1
32 # minimum temploral overlap (% of annotation duration) to count a detection window as containing a manua
33 minO = 0.5
34 # either specify channel(s) of interest, or leave as an empty list to analyze all available channels
35 chans = 1
36
37 # Calculations -------------------------
38
```

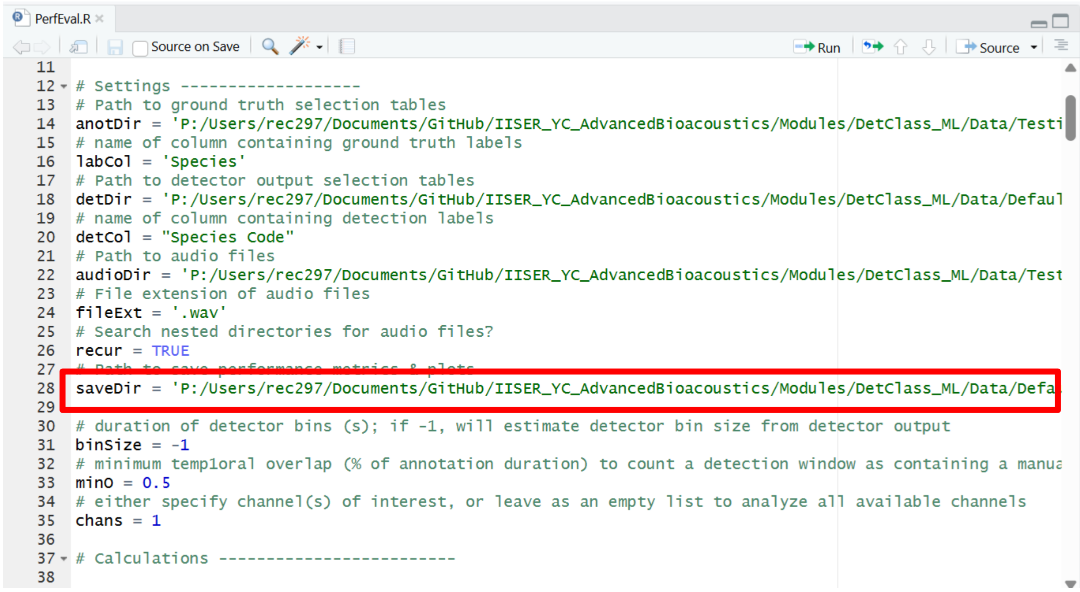
3. Update the `detCol` variable in line 20 to "Species Code"



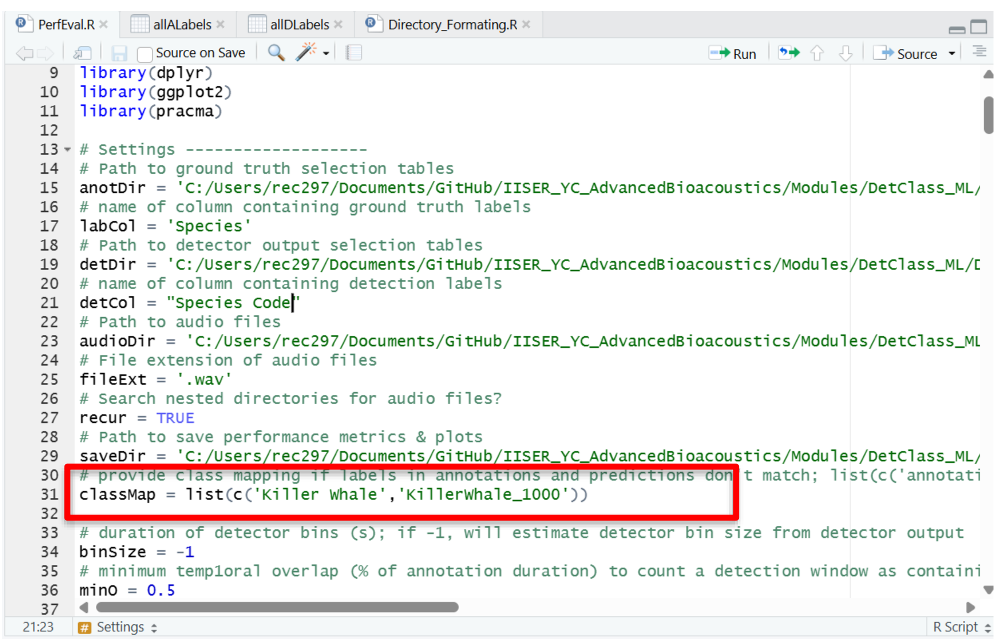
4. Update the `audioDir` variable path in line 22 to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing>> AudioFiles >> Marine*



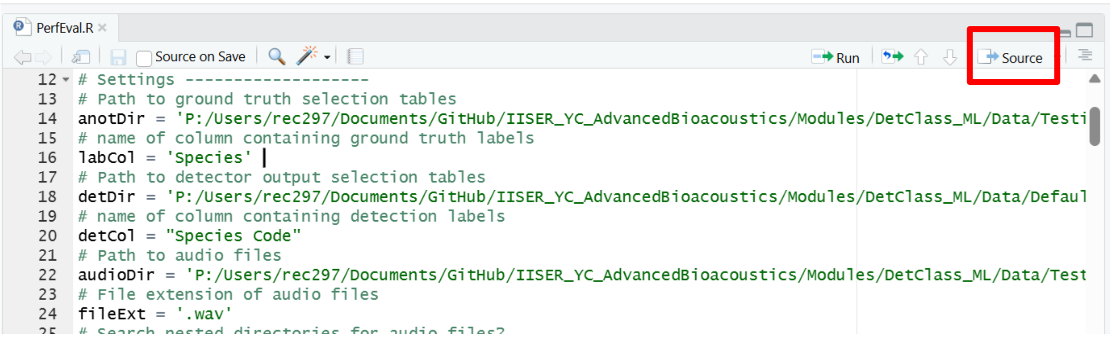
5. Update the `saveDir` variable path in line 28 to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> CustomModel*



6. Update `classMap` variable in line 31 to reflect name of training data folder your group used.



7. Run script by clicking ***Source*** button in upper right corner of script editor window



8. In Windows Explorer or FInder, navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> CustomModel* to view output figures.